FLOATING SOLAR PROJECT OF GREENAM ENERGY

Document Prepared by LGAI Technological Center S.A.

(Applus+ Certification)

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Summary:

LGAI Technological Center S.A. (Applus+ Certification) (Hereafter referred as Applus+ Certification) has been appointed by “EKI Energy Services Limited” to perform the validation and verification of the “Floating Solar Project of Greenam Energy” (VCS ID 4187[/1/])). The main purpose of this Validation and verification activity is to have an independent third-party assessment of the project design, monitoring system to ensure a thorough assessment of the project activity against the applicable VCS requirements.

The project activity is 24.7MWp (22 MWac) floating solar PV plant on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district, Tamil Nadu, India by Greenam Energy Private Limited (GEPL) which is a SPV floated by Southern Petroleum Industries Corporation Ltd (SPIC) and Greenstar Fertilizers (GFL) and AM International Holding Private Limited (AMIH). GEPL is jointly owned by SPIC Ltd (Southern Petrochemical Industries Corporation Limited), Greenstar Fertilizers Limited (GFL) and AM International Holding Private Limited (AMIH) which was identified based on the certificate of ownership as per rule 3 of Electricity Rule 2005. The generated electricity is transmitted through the Indian electricity grid to identified end users i. e. AMIH Group companies - Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL). A common power purchase agreement (PPA[/2/]) had been executed between GEPL and the off-takers i.e. SPIC and GFL which was further modified as energy wheeling agreement[/2/]. For the current crediting period, the annual estimated power generation from the project activity is approximately about 41,241 MWh which results in annual estimated average emission reductions of 38,376 tCO_2e/year and 268,629 tCO_2e over the 7 year crediting period. Over the 1st monitoring period from 12-January-2022 to 28-February-2023, the project has generated 36,856 MWh and claimed 36,174 tCO_{2e} emission reductions.

Validation objective: The objective of the validation is to confirm whether the project proponent has identified and presented information correctly in the PD relating to the baseline, estimated GHG emission reductions or net anthropogenic GHG removals, the monitoring plan and the crediting period using the valid version of the approved methodology and, where applicable, the approved standardized baseline and the other methodological regulatory documents that are applicable to the project activity.

Verification objective: The objective of the verification activity is to have an independent third party for the assessment of the project design, Actual ER sheet and to ensure a thorough assessment of the project activity against the applicable CDM and VCS requirements.

VVB has confirmed that the project activity conforms with;

- Baseline requirements of “AM0123 Renewable energy generation for captive use, version 01.0[/3/]”
- Monitoring requirements of “AM0123 Renewable energy generation for captive use, version 01.0[/3/]”
- CDM Validation and Verification Standard for project activities, version 03.0[/4/]
- VCS standard version 4.5[/4/]
- VCS program guide version 4.4[/4/]
- VCS validation and verification manual version 3.2[/4/]

VVB has checked the baseline and it was found to appropriately discussed and demonstrated. The project has discussed financial additionality and the project is additional which is confirmed. During the current monitoring period, project activity has undergone continued operation and no major breakdown had taken place.

The verification is carried out for the first monitoring period covered in the 1st crediting period (CP) from 12-January-2022 to 11-January-2029 which includes monitoring period from 12-January-2022 to 28-February-2023 (both days included). Total emission reductions achieved in this period are 36,174 tCO_2e.

A risk-based approach has been followed to perform this joint Validation and verification activity. In the course of joint Validation + verification, 07 Corrective Action requests (CARs) and 05 Clarification Requests (CLs), 00 Forward action request (FARs) were raised and successfully closed. The review of the revised VCS Project Description and Monitoring Report (PDMR) along with the supportive documents related to baseline and monitoring methodology; the subsequent background investigation, on-site visit have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to validate and verify the fulfilment of the stated criteria of VCS mechanism.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus + Certification) has been appointed by “EKI Energy Services Limited” to perform the Joint validation and verification of the project entitled “Floating Solar Project of Greenam Energy” by Greenam Energy Private Limited (Project Developer) under VCS standard version 4.5 and program guide version 4.4. The objective of this joint validation & verification activity is to have an independent third party assessment of the project design, baseline, additionality, emission reduction calculations and to ensure a thorough assessment of the project activity against the applicable CDM and VCS requirements. In particular;

- The project’s baseline is assessed against “AM0123: Renewable energy generation for captive use, version 01.0”
- The project’s monitoring plan is assessed against “AM0123: Renewable energy generation for captive use, version 01.0”
- VCS standard version 4.5
- VCS program guide version 4.4
- CDM Validation and Verification Standard for project activities, version 03.0
- Applicable tools and guidelines as per Clean Development Mechanism (CDM) and VERRA requirements

Validation & verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project, its registration with VERRA and its intended generation of estimated verified Carbon units (VCUs).

1.2 Scope and Criteria

The scope of the Joint Validation and verification is the independent and objective review of the Joint VCS Project Description & Monitoring Report (Joint PDMR). The Joint VCS PDMR was reviewed against the relevant criteria and decisions by the VCS, including the approved baseline and monitoring methodology. The validation and verification were based on the guidance given in the VCS program guide version 4.4 and VCS standard version 4.5.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the Joint VCS PDMR. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of Voluntary Carbon Units (VCU). The joint Validation and verification are not meant to provide any consulting to the project developer or participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design and monitoring report combined.

The only purpose of this joint Validation and verification is its usage during the registration /issuance process as a part of the VCS project cycle. Therefore, LGAI Technological Center S.A.

(Applus+ Certification) can't be held liable by any party for decisions made or not made based on the validation/verification opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions and Level of Assurance

The joint Validation and verification has been planned and organized to achieve a reasonable level of assurance as per the requirement of VCS.

1.4 Summary Description of the Project

The project activity involves generation of electricity by using the renewable energy source (solar energy) supplying to the Indian electricity grid. Thus, the project aims to displace electricity produced by power grid, which is dominated by fossil fuel power plants[1], harnessing solar energy. Therefore, the project reduces greenhouse gas emissions and thereby contributes to sustainable development.

VVB confirmed that, the 22 MWac solar power project is developed by Southern Petrochemical Industries Corporation Limited (SPIC), Greenstar fertilizers (GFL) and AM International Holding Private Limited (AMIH)); through a special purpose vehicle (SPV) company Greenam Energy Private Limited (GEPL). GEPL is the project developer/legal owner of the project activity and is jointly owned[2] by SPIC Ltd (Southern Petrochemical Industries Corporation Limited), Greenstar Fertilizers Limited (GFL) and AM International Holding Private Limited (AMIH). The generated electricity is fed to the grid and same is then consumed by Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL) who are legally bound by power purchase agreement GEPL with the for purchase of total electricity generated from the project activity. SPIC and GEPL are parent company of GEPL as then own 100% ownership in the GEPL as evident from energy wheeling agreement[/2/].

The project is located on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district, Tamil Nadu, India with the geographical Coordinates of 8°44'6"N to 78°7'44"E.

The project was commissioned on 12-January-2022 as confirmed based on commissioning certificate[/6/] and started sending the electricity into the grid. The local state electricity authority – Tamil Nadu Generation and Distribution Corporation (TANGEDO) based on quantum of electricity injected into the grid as recorded on the energy meter allocates equivalent power to the end users through signed tri party agreement.

PD has considered 12-January-2022 as the project start date which is appropriate. Assessment team checked the commissioning status of the project activity with the commissioning certificates and found correct.

[1] https://powermin.gov.in/en/content/power-sector-glance-all-india
[2] This has been confirmed based on certificate of ownership as per rule 3 of Electricity Rule 2005

No event observed during the current monitoring period which can alter or deviate from the methodology requirement.

The project lifetime is 25 years[3] as confirmed based on supportive weblink.

The project's technical details as checked based on submitted purchase orders/7/, commissioning certificates/6/, power purchase agreement/2/ and single line diagram and confirmed during the on-site visit are as follows:
Technical Specifications:

Components	Make
Modules (free issue)	Jinko solar
Inverters	Sungrow Inverters
Transformer	Power transformer – Hammond Power Solution Inverter Transformer – Essennar
VCB	ABB
MMS	Ceil & Terre
Module Cleaning	Considered manual cleaning with reservoir water
Weather Monitoring station and Sensors	Kipp and Zonen
SCADA	Trinity Touch
Remote Monitoring System (RMS)	Trinity Touch

Transformer details:

Current Req No	1. 1 No of Transformer (20-22 MVA) 2. 4 No's of Transformer (6.250 MVA * 3 No's, 3.250 MVA*1 No)
Future Expansion	NA
Details	Power Transformer 20/22 MVA *1 No 1. Ratio – 11 KV / 110 KV 2. Cooling – ONAN/ONAF 3. Winding – Copper 4. HV Connection – Bushing; LV Connection – Cable Box 5. HV Winding - Star, LV winding – Star 6. Vector Group – YNYN0 7. Impedance -10% Max Inverter Transformer 1. 6250MVA & 3250MVA – 4 No's 2. Ratio – 0.600 KV / 11 KV 3. Cooling – ONAN 4. Winding – Copper 5. HV Connection – Cable box; LV Connection – Cable Box 6. HV Winding - Delta, LV winding – Star 7. Vector Group – Dy11y11/ Dy11

Modules details:

[3] https://cercind.gov.in/2020/regulation/159_reg.pdf

Modules Details	1. 1500 V System 2. Make –Jinko Solar , Type: Mono PERC, Rated capacity -385 Wp 3. Qty – 65800 no's

Modules details:

Inverters Details	1. 1500 V System 2. Make –Sungrow , Type -Central Inverter 3. 3125 KVA@50deg, 1500 Vdc/600 Vac 4. Qty – 7no's

Below are the entities involved under project activity: -

The project legal owner is GEPL and its ownership is confirmed based on commissioning certificates, power purchase agreement. The Greenam Energy Private Limited (GEPL) - owner of solar plant and consumers Southern Petroleum Industries Corporation Ltd (SPIC) and Greenstar Fertilizers (GFL) are part of the same group i. e. AM international and GEPL is a SPV floated by SPIC and GFL., the same is verified from shareholding document and AM International website: https://aminternational.sg/businesses/green-solutions/.

Hence, section 1.5 of submitted PDMR provide details for all of the entities involved in the project which is checked and found appropriate.

Organization name	AM International Holding Private Limited
Role in the project	Project developer/owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Southern Petrochemical Industries Corporation Limited
Role in the project	Project developer/owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Greenstar Fertilizers Limited
Role in the project	Project developer/owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032

Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Greenam Energy Private Limited
Role in the project	Project legal owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

The other party involved is engaged in development of the project for carbon finance purpose only.

Organization name	EKI Energy Services Limited
Role in the project	Project Consultant
Contact person	Mr. Manish Dabkara
Title	MD & CEO
Address	Enking Embassy, Plot 48, Scheme 78 Part-2, Behind Vrindavan Hotel, Vijay Nagar, Indore-452010, Madhya Pradesh, India
Telephone	0731 428 9086
Email	registry@enkingint.org

PP has applied for renewable crediting period of 3 * 7 years. Thus, the first crediting period shall begin with the project start date i. e. 12-January-2022 and last until the end of 07 years period i.e. 11-January-2029.

The current monitoring period is from 12-January-2022 to 28-February-2023 (both days included). Emission reductions achieved in this period are 36,174 tCO_2e.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

Validation and Verification Scope:

The scope is defined as an independent and objective review of the Joint PDMR. The Joint PDMR is reviewed against the criteria stated in VCS standard version 4.5 and program guide version 4.4, approved CDM baseline and monitoring methodology AM0123: Renewable energy generation for captive use, version 01.0.

The joint Validation and verification is based on the requirements in the CDM Validation and Verification Standard (CDM VVS) for project activities version 03.0, and VCS program guide version 4.4 and VCS standard version 4.5.

The joint Validation and verification is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the combined project document and the Monitoring report.

Validation and Verification Process:

The project assessment is based on the CDM VVS for project activities version 03.0 and VCS standard version 4.5 and VCS program guide version 4.4 and is conducted using standard auditing techniques to assess the correctness of the information provided by the project participants. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

Once the project is received by the assessment team, the members of the assessment team carried out:

- A desk review of the Joint project design documentation and monitoring report;
- Onsite visit with project stakeholders;
- The resolution of outstanding issues and the issuance of the final Joint validation and verification report and opinion.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI Technological Center S.A. (Applus+ Certification) has conducted a discussion on each criterion by the assessment team, and the results from validating/verifying the identified criteria.

Appointment of the assessment team: -

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The four qualification levels for team members that are assigned by formal appointment rules are as presented below:

- Lead Auditor (LA)
- Technical Expert (TE)
- Financial Expert (FE)
- Technical Reviewer (TR)

The sectoral scope / technical area knowledge linked to the applied methodology/ies shall be covered by the assessment team.

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Mr. Deepak Pundlik	LA/TE	YES	YES	No	YES
Mr. Jitendra Mohan Singh	FE	YES	YES	YES	YES

Mr. Denny Xue	TR	YES	YES	YES	NA

The complete list of CVs is included as Appendix 3 of this report.

Document Review: -

The Joint VCS PDMR submitted by the PD was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

Onsite Visit: -

An Onsite inspection is conducted by LGAI Technological Center S.A. (Applus+ Certification) assessment team who performed visual inspection of project site with project stakeholders to confirm project information and to resolve issues identified in the document review. The details are provided in this report in the below sections.

Resolution of Clarification and Correction Action Request: -

The objective of this phase of the joint validation and verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification) positive conclusion on the Joint PDMR. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communication between the PD and Applus+ Certification to guarantee the transparency of the validation/verification process, the concerns raised and responses given are summarized below in the appendix 2.

The most recent Joint PDMR submitted by project developer serves as the basis for the final assessment presented. Additional changes to the project during the joint validation and verification process are not considered to be significant with respect to the main VCS objectives i. e. the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Internal quality Control: -

As final step of a joint Validation and verification of the final documentation including the final Joint Validation and verification report and the checklist, have to undergo an internal quality control by the technical review committee, i.e., each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of Interest.

After confirmation of the project participants, the positive validation/verification opinion and relevant documents are submitted to the VCS secretariat through the VCS web-platform.

2.2 Document Review

The details of the document observed during the joint Validation and verification process are listed below in appendix 1 of this report.

2.3 Interviews

The names of the persons interviewed during site visit is given below;

Date of Interview: 29-June-2023 to 30-June-2023		
Sr. No.	Name of Persons	Role/Designation
1	Mr. Vishal Vijay	Site in-charge, EKI Energy Services Limited
2	Mr. Surjit Chanda	EKI Energy Services Limited
3	Mr. Suraj Prajapati	O & M head, EKI Energy Services Limited
4	Mr. Nikhil Sukhdaliya	Local stakeholder

2.4 Site Visits

Duration of onsite audit: 29-June-2023 and 30-June-2023				
No.	**Activity performed on-site**	**Site location**	**Date**	**Team member**
1.	Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, technical description of the project and Monitoring. Assessment team also checked that whether the monitoring plan as described in the VCS PDMR is actually practised onsite. Also, assessment team checked any change in host country criteria which may affect the baseline of the project activity.	Muthiahpuram village, Thoothukudi distric, Tamil Nadu	29-June-2023 and 30-June-2023	Mr. Deepak Pundlik

2.5 Resolution of Findings

The objective of this phase of the joint Validation and verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on JPDMR. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the project developer and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation/verification process, the concerns raised and responses given are summarized below in the Appendix 2.

The final Joint PDMR submitted by project developer serves as the basis for the final assessment presented. Additional changes to the project during the joint validation and verification process are not considered to be significant with respect to the main VCS objectives which are reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project design document and Monitoring report	-	02	-
Description of project activity	03	02	-
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	01	-	-
- Deviation from methodology	-	-	-
- Clarification on applicability of methodology, tool and/or standardized baseline	-	-	-
- Demonstration of additionality	-	01	-
- Emission reductions	-	01	-
- Monitoring plan	-		-
- Stakeholder's consultation process	-	01	-
- Public comments	-	-	-
Others (SDGs)	01	-	-
Total (Validation+ Verification)	05	07	00

The list of findings and their resolution is presented in Appendix 2 of this report.

2.5.1 Forward Action Requests

No FAR is raised during the current joint Validation and verification process.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project:

The project activity involves generation of electricity by using the renewable energy source (solar energy) supplying to the Indian electricity grid and then further use of equivalent amount of electricity by the identified end users through contractual agreement between project developer, state electricity board and end users. Thus, the project aims to displace electricity produced by fossil fuel power plants harnessing solar energy. Therefore, the project reduces greenhouse gas emissions and thereby contributes to sustainable development.

Project location:

The current project is a floating solar plant installed on water reservoirs and operated by Greenam Energy Private Limited (GEPL). The project activity is 24.7MWp (22 MWac) floating solar PV plant on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district, Tamil Nadu, India. The generated electricity is fed into grid and equivalent amount is used by the end users i. e. Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL) through contractual arrangement with GEPL.

VVB has confirmed that the GEPL is jointly owned by SPIC, GFL and AM International which can be identified based on certificate of ownership as per rule 3 of Electricity Rule 2005[4]. Thus, this meets the criteria as mentioned in AM0123: Renewable energy generation for captive use which is also confirmed based on energy wheeling agreement between GEPL (project developer) and SPIC and GFL (end users) wherein all 3 are owned by AM International[5].

Thus, VVB confirmed that project type is renewable energy generation through solar power, subsequent use of the same for captive purpose and same is eligible under the VCS for claiming the emission reductions.

Project design:

The proposed project is a single location project which involves installation and commissioning of 22 MWac solar power plant. The technical specifications of the project activity are presented in JPD&MR which were checked against the submitted technical datasheet and confirmed during on-site visit and are as below:

Components	**Make**
Modules (free issue)	Jinko solar
Inverters	Sungrow Inverters
Transformer	Power transformer – Hammond Power Solution Inverter Transformer – Essennar
VCB	ABB
MMS	Ceil & Terre
Module Cleaning	Considered manual cleaning with reservoir water
Weather Monitoring station and Sensors	Kipp and Zonen
SCADA	Trinity Touch
Remote Monitoring System (RMS)	Trinity Touch
Transformers	
Current Req No	1. 1 No of Transformer (20-22 MVA) 2. 4 No's of Transformer (6.250 MVA * 3 No's, 3.250 MVA*1 No)
Future Expansion	NA

[4] https://cea.nic.in/wp-content/uploads/legal_affairs/2020/09/Electricity-rules-2005.pdf
[5] https://aminternational.sg/businesses/green-solutions/ and https://aminternational.sg/businesses/fertilizers-supply-chain/

Details	Power Transformer 8. 20/22 MVA *1 No 9. Ratio – 11 KV / 110 KV 10. Cooling – ONAN/ONAF 11. Winding – Copper 12. HV Connection – Bushing; LV Connection – Cable Box 13. HV Winding - Star, LV winding – Star 14. Vector Group – YNYN0 15. Impedance -10% Max Inverter Transformer 8. 6250MVA & 3250MVA – 4 No's 9. Ratio – 0.600 KV / 11 KV 10. Cooling – ONAN 11. Winding – Copper 12. HV Connection – Cable box; LV Connection – Cable Box 13. HV Winding - Delta, LV winding – Star 14. Vector Group – Dy11y11/ Dy11
Modules Details	1. 1500 V System 2. Make –Jinko Solar , Type: Mono PERC, Rated capacity - 385 Wp 3. Qty – 65800 no's
Inverters Details	1. 1500 V System 2. Make –Sungrow , Type -Central Inverter 3. 3125 KVA@50deg, 1500 Vdc/600 Vac 4. Qty – 7no's

The project was commissioned on 12-January-2022 (Same is considered as Project start date).

Assessment team checked the Commissioning status of the project activity with the commissioning certificates and found correct. No event observed during the current monitoring period which can alter or deviate from the methodology requirement.

Baseline Scenario assessment:

As per the paragraph 21 (a) of applicable approved methodology AM0123 version 1.0, "S1: No investment is undertaken by the project participant to meet the electricity demand of the captive consumer, and electricity delivered to the captive consumer would have been generated by the operation of grid-connected power plants and by the addition of new generation sources;" More details on the same are provided in section 3.4.4 below.

The project baseline is the "Grid-connected electricity generation". It complies with the current legal framework of the host country - India. There are no additional laws that came into force that has an impact on the project activity and the project activity is still in line with the available law and regulations.

Project crediting period:

PP has applied for renewable crediting period of 3 * 7 years. Thus, the first crediting period shall begin with the project start date i. e. 12-January-2022 and last until the end of 07 years period i.e. 11-January-2029.

The current monitoring period is from 12-January-2022 to 28-February-2023 (both days included). Total emission reductions achieved in this period are 36,174 tCO_2e.

The estimated lifetime of the project activity is considered as 25 years according to the limited warranty provided by supplier of solar PV modules.

Project proponent and other entities involved in the project:

Assessment team checked during onsite visit and confirms that below are the entities involved under project activity: -

Organization name	AM International Holding Private Limited
Role in the project	Project developer/owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Southern Petroleum Industries Corporation Ltd
Role in the project	Project developer
Contact person	P.Senthil Nayagam
Title	Director
Address	"SPIC HOUSE", 88 Mount Road, Guindy, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Greenstar Fertilizers Limited
Role in the project	Project developer/owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Greenam Energy Private Limited
Role in the project	Project legal owner
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED, "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032

Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	EKI Energy Services Limited
Role in the project	Project Consultant
Contact person	Mr. Manish Dabkara
Title	MD & CEO
Address	Enking Embassy, Plot 48, Scheme 78 Part-2, Behind Vrindavan Hotel, Vijay Nagar, Indore-452010, Madhya Pradesh, India
Telephone	0731 428 9086
Email	registry@enkingint.org

Project ownership is with AM International Holdings Private Limited which owns all the other three companies i. e. SPIC, GFL and GEPL[6]. GEPL is the company which legal owns and operates the solar plant. The ownership was confirmed through energy wheeling agreement/2/..

Project start Date: Project start date is 12-January-2022 since project has started power generation to feed into Indian electricity grid and generated the first GHG emission reduction as per the definition in VCS Standard version 4.5.

Assessment team checked the documents/6/ and details provided by the PP and start date found correct.

Project scale & estimated GHG Emissions Reductions or Removals:

Assessment team confirms that the project involves setting up the project activity of 22 MW(AC) installed capacity.

As per VCS standard version 4.5, the projects are classified as follows:

1) Projects: Less than or equal to 300,000 tonnes of CO_2e per year

2) Large Projects: Greater than 300,000 tonnes of CO_2e per year

The project activity has less than 300,000 tCO_2e Emission reductions per year, hence the project activity is classified as "Projects".

Project Scale	
Project	√
Large project	

[6] https://aminternational.sg/businesses/fertilizers-supply-chain/ and https://aminternational.sg/businesses/green-solutions/

Year	Estimated GHG emission reductions or removals (tCO_2e)[7]
Year 1 (12-January-2022 to 31-December-2022)	37,614
Year 2 (01-January-2023 to 31-December-2023)	38,726
Year 3 (01-January-2024 to 31-December-2024)	38,559
Year 4 (01-January-2025 to 31-December-2025)	38,394
Year 5 (01-January-2026 to 31-December-2026)	38,229
Year 6 (01-January-2027 to 31-December-2027)	38,064
Year 7.a from (01-January-2028 to 31-December-2028)	37,900
Year 7.b from (01-January-2029 to 11-January-2029)	1,142
Total estimated ERs	268,629
Total number of crediting years	07
Average annual ERs	38,376

Condition prior to project initiation: -

Assessment team during the desk review and onsite assessment confirms that the project implementation by Greenam Energy and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction.

The baseline as described in section 3.4.4 of this report will continue to be the baseline in the absence of project activity.

Project compliance with applicable laws, statutes and other regulatory frameworks:

VVB has confirmed that the project is in compliance with applicable laws, statues, and other regulatory frameworks as below:

- Electricity Act 2003 (May 2007 Amendment)[8]
- National Electricity policy 2005[9]
- The Electricity (Supply) Act, 1948[10]

[7] The value is considering the degradation factor for the solar panels

[8] https://cercind.gov.in/Act-with-amendment.pdf
[9] https://powermin.gov.in/en/content/national-electricity-policy
[10] https://cercind.gov.in/ElectSupplyAct1948.pdf

- Environmental (Protection) Act, 1986[11] and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006[12] and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)[13]
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003[14].
- Noise Pollution (Regulation and Control) Amendment Rules, 2017[15]
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016[16] and amended in November, 2021.
- E-waste (Management and Handling) Rules, 2020[17].
- Plastics Waste Management Rules, 2016 amended in 2021[18]
- Batteries (management and handling) rules 2020[19].

Thus, assessment team confirms that the project activity follows the National and local law and regulation of the host country.

Project ownership:

Greenam Energy Private Limited (GEPL) is project developer (PD) of project activity and they have the legal right to control and operate the project activity. VVB further noted that GEPL is a SPV floated by SPIC, GFL and AMIHPL for this particular project and they have 100% ownership of GEPL.

The project ownership has been checked by the VVB through commissioning certificate, power purchase agreement and energy wheeling agreement along with auditor's certificate.

Participation under other GHG programs:

VVB has checked other GHG programs and confirm that the project is not registered (or seeking registration) under other GHG programs except for Global Carbon Council (GCC)[20]. VVB further confirms that since, the registration deadline for GCC project is now over, the project application for registration is no more valid with GCC and PP has not applied again for the same.

VVB has checked other GHG programs such as CDM, GS4GG, GCC , I-Rec and confirm that project was not rejected by other GHG program.

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[11] https://parivesh.nic.in/writereaddata/ENV/eprotect_act_1986.pdf
[12] http://www.environmentwb.gov.in/pdf/EIA%20Notification,%202006.pdf
[13] https://cpcb.nic.in/air-pollution/#:~:text=The%20Air%20(Prevention%20and%20Control,of%20air%20pollution%20in%20India.
[14] https://cpcb.nic.in/water-pollution/#:~:text=The%20Water%20(Prevention%20and%20Control%20of%20Pollution)%20Cess%20Act%20was,certain%20types%20of%20industrial%20activities.
[15] https://upload.indiacode.nic.in/showfile?actid=AC_CEN_16_18_00011_198629_1517807327582&type=rule&filename=2555%20dated%2010.08.2017%20noise%20pollution%20.pdf
[16] https://cpcb.nic.in/rules/
[17] https://cpcb.nic.in/e-waste/
[18] https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1745433#:~:text=and%20Climate%20Change-,Government%20notifies%20the%20Plastic%20Waste%20Management%20Amendment%20Rules%2C%202021%2C%20prohibiting,from%20the%2031st%20December%2C%202022.
[19] https://www.eqmagpro.com/wp-content/uploads/2020/02/Battery-Waste-Management-Rules-2020-draft.pdf
[20] https://projects.globalcarboncouncil.com/project/670

Participation under other Emission Trading programs & other binding limits / other form of environmental credit sought or received and eligible to be sought or received: -

Assessment team confirms that the Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. Further, Declaration/10/ in effect of the same has been submitted by project proponent to audit team and found to be correct. Thus, it is concluded that the project activity not involved on other Emissions trading programs and other binding limits.

Leakage Management for AFOLU project: -

Not applicable to the project activity

The project activity is generation of the electricity using solar modules in the Rajasthan state, Hence, the VVB verifier has confirmed that there is no emission due to supply chain.

Commercially Sensitive Information: -

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation and verification.

Further, in line with VCS Standard version 4.5, para 3.16.1, "The project proponent must demonstrate that a project contributes to at least three SDGs by the end of the first monitoring period, and in each subsequent monitoring period."

The project is expected to contribute 3 SDGs which are SDG 7, 8 and 13 for the current crediting period.

SDG 7 Energy: The project contributes SDG Target 7.2 "By 2030, increase substantially the share of renewable energy in the global energy mix" and Indicator 7.2.1 by the utilization of Solar power as a renewable energy source.

SDG 8 Decent Work and Economic Growth: During the operation phase of the project, job opportunities are created. Therefore, the project contributes to SDG Indicator by generating employment of local men and women. PP has targeted to employ 10 persons.

SDG 13 Climate Change: The project produces clean renewable energy by diminishing CO_2 emissions. Therefore, it contributes SDG Target 13.0 "Tonnes of greenhouse gas emissions avoided or removed".

VVB have assessed the SDG 7.2, SDG 8 and SDG 13. and supporting documents shared by PP and found values correct as provided in Section 1.17.2 of Joint PDMR.

CL 05 was raised and successfully closed.

Hence, Assessment team confirms that the purpose of this project activity is to reduce GHG emission Solar Power Plant of 22 MW(AC) in Tamil Nadu by Greenam Energy Private Limited. Same is confirmed through document review and onsite visit with the project site personnel. Final Joint PDMR also reflects the same. Thus, found acceptable as Joint PDMR is accurate and provided complete project information.

VVB further confirms that the description in the Final Joint PDMR is accurate, complete, and provides an understanding of the nature of the project. The project has been implemented as described in the project description, VVB confirms that the project is likely to achieve estimated GHG emission reduction or removals however the actual results have varied since the initial estimates wee based on assumptions that are subject to change.

3.2 Participation under Other GHG Programs

The project has neither been registered nor seeking registration under any other GHG programs except for Global Carbon Standard (GCC). PD had applied for registration with GCC however, it was cancelled by PD and the project is seeking registration only in VCS program. Further, declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

1. http://cdm.unfccc.int/
2. http://www.goldstandard.org/
3. https://www.globalcarboncouncil.com/

PD had applied for GCC registry but has decided not to conclude the validation with the same and continue the project under VERRA.

Rejection by other GHG program: -

The Project is not rejected by other GHG programs. A declaration for the same is checked and found correct by the assessment team. Also, the assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

1. http://cdm.unfccc.int/
2. http://www.goldstandard.org/
3. https://www.globalcarboncouncil.com/

The Project does not intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

3.3 Safeguards

3.3.1 No Net Harm

The project activity promotes environmental and socio-economic well-being as it results in zero GHG emissions due to implementation and operation of clean, renewable energy technology for electricity generation. VVB confirms that solar project does not cause any net harm and no negative environmental and socio-economic impacts are being identified by PP which is acceptable.

3.3.2 Local Stakeholder Consultation

The Joint PDMR section 2.2 contains a detailed explanation of the local stakeholder consultation process conducted. Stakeholders were defined as individuals who are impacted by project activities and those whose actions directly or indirectly affect the project.

The stakeholder consultation meeting details are as follows:

The dates of the Stakeholder Meetings are as follows:

Date of invitation – 02-March-2022

Date of Meeting – 11-March-2022

Location of Meeting – Plant Office, Greenam Energy Private Limited

Village representatives like village head, their representatives along with the local household owners were invited to the meeting. The purpose of the meeting was to explore any concerns that the stakeholders may have had about the project's environmental and social implications. Throughout the stakeholders meeting, opinions and recommendations were welcomed.

The assessment team conducted stakeholder interviews on-site. The validation team confirmed by the stakeholder's responses that the stakeholder consultation process had been carried out in accordance with the joint PDMR. The stakeholder acknowledged that they received an invitation/11/ to attend the meeting. The invitation procedure specified in the joint PDMR was determined to be consistent with this claim. When the assessment team asked stakeholders about the grievance reporting mechanism, they confirmed that they had been informed during the stakeholder consultation process. A copy of the grievance register confirms that there were no complaints reported during the current monitoring period. A few questions were raised by the stakeholders which were satisfactorily addressed by the PP team.

PP has noted feedback received during the meeting in the VCS PDMR. VVB noted that stakeholder inputs have no effect on the project design and no changes were necessary to project design. VVB further confirms that inputs received were appropriately addressed by the PP.

CAR 04 was raised and closed successfully. Kindly refer appendix 3 for more information.

3.3.3 Environmental Impact

Solar PV is one of the cleanest sources of renewable energy, with no associated emissions and waste products. In India, Environmental Impact Assessment (EIA) of solar power project is not an essential regulatory requirement. The project activity has no significant impact on the environment.

As The guidelines on Environmental Impact Assessment have been published by Ministry of Environment, Forests and Climate Change (MoEF&CC), Government of India (GOI) under Environmental Impact Assessment notification 14/09/2006[21]. Further amendments to the notification have been done on 14/07/2018, the Solar Power projects are not listed in any of the categories of the schedule, hence, No EIA required as per host country legislation.

The assessment team has checked the provided regulation and reference for the same and confirm that there is no mandatory legal requirement for carrying out an environmental impact assessment in the host country.

3.3.4 Public Comments

No comments were received during the webhosting period 24-March-2023 to 23-April-2023. The details were checked by the assessment team and found correct.

3.3.5 AFOLU-Specific Safeguards

The project activity is not AFOLU project; Hence this section is not required.

[21] http://www.environmentwb.gov.in/pdf/EIA%20Notification,%202006.pdf

3.4 Application of Methodology

3.4.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Methodology:
Methodology : AM0123: Renewable energy generation for captive use , version 1.0
Project Type : Type-I
Sectoral scopes : 1

The methodology refers to following CDM tools:

- Tool 02 - Combined tool to identify the baseline scenario and demonstrate additionality, version 07.0[22]
- Tool 05 - Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
- Tool 07 - Tool to calculate the emission factor for an electricity system, version 07.0[23]
- Tool 24 – Common practice, version 03.1[24]
- Tool 27- Investment analysis, version 12.0[25]

As per the applied methodology, tool 03 and tool 05 are to be used wherever applicable. However, for the current project activity these tools are not required as the project involves solar power based electricity generation and does not have project/leakage emissions from fossil fuel usage as well as baseline, project , leakage emissions from electricity consumption and monitoring of electricity generation is not required.

3.4.2 Applicability

The choice of methodology AM0123, version 1.0 is justified as the project activity meets the relevant applicability criteria:

The Solar Power Plant Project involves installation of a 22 MW(AC) grid connected renewable electricity generation plant for captive usage.

The applicability conditions of the methodology as mentioned below were checked and assessment team confirmed that they are valid as confirmed based on the submitted documentary evidences and observations/interviews during the site visit:

[22] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf
[23] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf
[24] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf
[25] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf

Paragraph no.	Applicability Conditions, AM0123 (Ver 01.0)	Justification of eligibility	VVB assessment
2	This methodology applies to a greenfield renewable energy power generation project activities supplying electricity to the captive consumer. Where necessary the greenfield power plant may be integrated with battery energy storage systems (BESS).	Floating Solar project of Greenam Energy (henceforth called “the project”) is a greenfield renewable energy power generation project (floating solar PV plant), supplying electricity to the captive consumer. Hence, the project activity meets the applicability condition of the methodology.	The proposed project activity is solar power based electricity generation supplying electricity to the captive consumers through grid interface through wheeling. The project does not involve BESS. VVB has confirmed this during on-site visit and based on commissioning certificate, power purchase agreement, energy wheeling agreement. Hence this criterion is applicable.
3	This methodology is applicable to renewable energy power generation project activities that install a greenfield power plant supplying electricity to the captive consumer via: 1. A grid interface through a wheeling; or 2. A dedicated electricity transmission and distribution line.	The project is implemented by Greenam Energy Private Limited (GEPL) which is a SPV formed by SPIC who is the project participant and energy generated shall be supplied to Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL), via a grid interface through a wheeling. Hence, the project activity meets the applicability condition of the methodology.	The proposed project activity is solar power based electricity generation supplying electricity to the captive consumers (SPIC and GFL) through grid interface through wheeling. VVB has confirmed this during on-site visit and based on commissioning certificate, power purchase agreement, energy wheeling agreement. Hence this criterion is applicable.

4	In case the project activity integrates a BESS, the methodology is applicable only to project activities that integrate greenfield BESS with a greenfield renewable energy power plant.	Project activity is greenfield power project and not integrated with BESS, hence para 4 is not relevant to the project activity.	The proposed project activity is solar power based electricity generation supplying electricity to the captive consumers through grid interface through wheeling. The project does not involve BESS. VVB has confirmed this during on-site visit and based on commissioning certificate, power purchase agreement, energy wheeling agreement. Hence this criterion is not applicable.
5	The methodology is applicable under the following conditions: (a) In the pre-project scenario, the captive consumer does not source electricity from a renewable energy source; (b) The renewable energy producer and captive consumption facility shall be owned by the same project participant; (c) The project activity may include a renewable energy power plant/unit of one of the following types: solar power plant/unit, wind power plant/unit, and/or hydro power plant/unit with or without a reservoir. (d) The greenfield project power plant should, at the time of the start date	(a) In the pre-project scenario, the captive consumer does not source electricity from a renewable energy source; Hence, the project activity meets the applicability condition of the methodology. (b) Greenam Energy Private Limited (GEPL), Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL) are group companies of AM International Holdings Pte. Limited. Hence, it can be concluded that the renewable energy producer (GEPL) and captive consumption facility (SPIC & GFL) are owned by the same project participant. Hence, the project activity meets the applicability	The proposed project activity is solar power based electricity generation supplying electricity to the captive consumers through grid interface through wheeling. VVB has confirmed this during on-site visit and based on commissioning certificate, power purchase agreement, energy wheeling agreement. Hence this criterion is applicable. (a) the electricity bills of SPIC and GFL and on-site interviews have confirmed in the baseline scenario, the captive consumers were not sourcing electricity from a renewable energy source (b) The same project participant i. e. SPIC owns the renewable energy producer and captive

	of the project activity, be contractually bound to supply electricity to a captive consumer at least for the entire duration of the first crediting period. (e) The project renewable energy plant(s) shall not export more than 10 per cent on a yearly basis of its generation to a grid. This condition shall be met over the crediting period; (f) In case of integration of BESS as per paragraph 4 above, the project participants shall demonstrate that the BESS was an integral part of the design of the project activity (e.g., by referring to feasibility studies or investment decision documents); (g) The BESS should be charged with electricity generated from the project renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a backup generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the	condition of the methodology. (c) The project activity involves installation of the "greenfield power project" generating electricity using solar plant. Hence, the project activity meets the applicability condition of the methodology. (d) Project activity is contractually bound before implementation via a long term common power purchase agreement (PPA) executed between GEPL and the off-takers, namely SPIC and GFL. Hence, the project activity meets the applicability condition of the methodology. (e) Project activity supplies 100% of electricity generated to captive consumer and this condition will be verified for the entire crediting period. Hence, the project activity meets the applicability condition of the methodology. (f) Project activity is greenfield power project and not integrated with BESS, hence this condition is not relevant to the project activity.	consumption facility which is confirmed based on energy wheeling agreement, electricity act, 2005 and auditor's certificate regarding the ownership details of Greenam Energy Pvt. Ltd. (c) The project involves solar power plant as confirmed during on-site visit. (d) The power purchase agreement and subsequent energy wheeling agreement with captive consumers is valid for entire duration of 1st crediting period is checked by VVB. (e) The power purchase agreement and subsequent energy wheeling agreement with captive consumers is valid for 100% consumption of generated electricity for entire duration of 1st crediting period is checked by VVB (f) The project is does not involve integration of BESS as confirmed during on-site visit by VVB. (g) The project is does not involve integration of

	requirements under section 5.4.3 below. The charging using the grid or backup generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant on an annual basis. During the time periods (e.g., days(s), week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant, for the concerned periods of the monitoring period, shall consider baseline emissions as zero, however, they shall account the project emissions as per requirements under section 5.4.3. Below.	(g) Project activity is greenfield power project and not integrated with BESS, hence this condition is not relevant to the project activity.	BESS as confirmed during on-site visit by VVB.
6.	VVB has checked that paragraph 6 and 7 are applicable to hydro power plants. The current project is solar based renewable energy generation project and hence paragraph 6 and 7 are not applicable for the project activity.		
7			
8	In addition, the applicability conditions included in the tools referred to below apply.	Applicability conditions of the tools included has been justified below.	PP has discussed applicability conditions of tools in the JPD&MR which are assessed below and confirmed.

Tool 2: "Combined tool to identify the baseline scenario and demonstrate additionality (version 07)	
Applicability Conditions	**VVB assessment**
The tool is applicable to all types of proposed project activities. However, in some cases, methodologies referring to this tool may require adjustments or additional explanations as per the guidance in the respective methodologies. This could include, inter alia, a listing of relevant	VVB noted that the applied methodology AM0123 version 1.0 paragraph 11(a) refers to Tool 2 hence this tool is applicable to the current project activity.

Tool 2: "Combined tool to identify the baseline scenario and demonstrate additionality (version 07)	
Applicability Conditions	**VVB assessment**
alternative scenarios that should be considered in Step 1, any relevant types of barriers other than those presented in this tool and guidance on how common practice should be established.	

Tool 05 - Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0	
Applicability Conditions	**VVB assessment**
If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	The proposed project activity is solar power based electricity generation supplying electricity to the captive consumers through grid interface through wheeling. In the absence of the project activity the captive consumers were purchasing the electricity from the gird only as confirmed during on-site visit and based on electricity bills submitted. Hence this criterion scenario (A) is applicable.
This tool can be referred to in methodologies to provide procedures to monitor amount of	The proposed project activity is solar power based electricity generation supplying

Tool 05 - Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0	
Applicability Conditions	**VVB assessment**
electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities	electricity to the captive consumers through grid interface through wheeling. VVB has confirmed this during on-site visit and based on commissioning certificate, power purchase agreement, energy wheeling agreement. Hence this criterion scenario II is applicable.
This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	The proposed project activity is solar power based electricity generation supplying electricity to the captive consumers through grid interface through wheeling. In the absence of the project activity the captive consumers were purchasing the electricity from the gird only as confirmed during on-site visit and based on electricity bills submitted. Hence this criterion is applicable.

Tool 07: Tool to calculate the emission factor for an electricity system (Version 07.0)	
Applicability conditions	**VVB assessment**
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	PD has used this tool to calculate values of OM, BM and CM in section B.6.1 of Joint PDMR for calculating of the baseline emissions which is checked and confirmed.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II.a and option IIb. If option II.a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the	The project is greenfield solar power project connected to electricity gird and hence VVB accepts that this condition is applicable.

grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in India, a Non-Annex I country. Therefore, this condition is not applicable to the project activity.
Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The project activity is a green field solar power plant and hence the condition of biofuel emission factor is not applicable.

Tool 24: Common practice (Version 03.1)	
Applicability Conditions	**Justification of eligibility**
This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality	The project activity is applying the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied approved baseline and monitoring methodology dose not define approaches for the conduction of the common practice test.

Tool 27: Investment analysis (Version 12.0)	
Applicability Conditions	**Justification of eligibility**
Para 2 - This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario. – Proposed project activity is "Tool for the demonstration and assessment of additionality"	The project activity applies Tool for the demonstration and assessment of additionality.
Para 3 - In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. – Methodology refers "General guidelines for SSC CDM methodologies, information on additionality (attachment A to Appendix B)	The applied approved baseline and monitoring methodology contains requirements for the investment analysis that are not different from those described in this methodological tool.

VVB confirms that the methodology is applicable to the project as the project activity is a grid connected greenfield Solar plant project and generated electricity is being wheeled through grid for the captive use.

In addition, for the emission factors, that were used to calculate estimated emission reductions, CEA database version 18 was used.

PP has explained the scenarios and applicability of the baseline methodology appropriately in the Joint PDMR and found correct, LGAI Technological Center S.A. (Applus+ Certification) confirms that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology is applicable to the project activity.

Project Boundary

3.4.3 As per paragraph 18 of applicable approved methodology, “The spatial extent of the project boundary includes the project power plant/unit and or all power plants/units connected physically to the electricity system that the CDM project power plant is connected to, the captive consumer that receives electricity generated by the project activity via wheeling, or through a dedicated electricity transmission and distribution line”. Thus, the project boundary for the proposed VCS project activity consists of Solar PV installation, connected substation, metering equipment’s, Indian electricity grid and the end users.

Table: Emissions sources included in or excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO_2	Yes	Main emission source: Fossil fuels fired for electricity generation cause CO_2 emissions. It is included to baseline calculation to find the displaced amount by the project activity.
		CH_4	No	Minor emission source. Excluded for simplification. This is conservative
		N_2O	No	Minor emission source. Excluded for simplification. This is conservative
3.4.4 Project	Emissions of CH_4 from the reservoir	CO_2	No	Minor emission source. The project activity will not create a new reservoir. Therefore, there will not be any project emission. This is conservative
		CH_4	No	
		N_2O	No	

Baseline Scenario

As per the paragraph 21 of applicable approved methodology AM0123 version 1.0, PP shall identify alternatives for baseline from para 21 (a) to (f) i. e. S1 to S6.

PP has selected following two plausible baseline scenarios which is found appropriate.

"S1: No investment is undertaken by the project participant to meet the electricity demand of the captive consumer, and electricity delivered to the captive consumer would have been generated by the operation of grid-connected power plants and by the addition of new generation sources;"

"S4; The project participant makes an investment to meet the electricity demand of a captive consumer through the construction of captive power plant using renewable power generation technologies, i.e., the proposed project activity is undertaken without being registered as a CDM project activity."

VVB during on-site visit has confirmed that captive consumers were using the electricity from operation of grid connected power plants and there was no investment undertaken by PP in the baseline. Hence, option (a) – scenario S1 is found appropriate. Scenario S4 i. e. without being registered as carbon (CDM) project is not feasible for PP which is detailed in section 3.5 of JPD&MR and assessment of the same is presented in section 3.4.5 below.

Thus, in line with para 22 of the applied methodology, scenario S1 is the economically most attractive alternative scenario as per step 3 of tool 02 which is discussed in section 3.5 of JPD&MR and section 3.4.5 of this report.

For calculation of baseline emission factor, as per paragraph 40 (a), Tool 07 is used and same is found acceptable.

Determination of Grid Emission Factor ($EF_{grid,CM,y}$)

The project participant used the "Tool to calculate the emission factor for an electricity system" to determine the emission coefficient as per 19 (a) of tool 05 "Tool to calculate the emission factor for an electricity system" states that electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations. In this case the Combined Margin (weighted average of Simple Operating Margin and Build Margin) is estimated based on three-year average (2019-20, 2020-21 & 2021-22) of Simple Operating Margin and Build Margin of current year (2021-22) is in line with steps of "Tool to calculate the emission factor for an electricity system", version 07.0. Both the value of Simple Operating Margin and Build Margin are selected under ex-ante approach. The grid boundary with respect to the connected grid is Indian national electricity grid.

In accordance with "Tool to calculate the emission factor for an electricity system" Dispatch Data Analysis" is the first methodological choice out of four options of calculating OM emission factor. Nevertheless, the "Dispatch data analysis operating margin" is ruled out in India due to lack of necessary dispatch data of the grids. The same fact is also considered by the Central Electricity Authority (Ref the user guide for CO_2 Baseline Database for the Indian Power Sector version 18.0, Sept'2022[26]).

[26] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

Out of other 3 options of calculating OM, Project participant have rightly selected simple OM emission factor calculation as the share of low cost / must run resources of the selected grid over the five most recent years (2017-18, 2018-19, 2019-20, 2020-21 & 2021-22) is less than 50% of the gross grid generation. For wind projects, “Tool to calculate the emission factor for an electricity system” allows the usage of the default weights are as follows: W_{OM}=0.75 and W_{BM} = 0.25. Using the above values, the combined margin emission factor is valued at 0.9310 tCO_2/MWh.

The calculation of $EF_{grid,CM,y}$ is current and publicly available and published by the Central Electricity Authority on its web-site. The assessment team is convinced of the result of the emission factor calculation. It is deemed to be adequate and transparent.

The assessment team confirms the following;

- All assumptions and data used by the project participants are listed in the Joint PDMR, including their references and sources;
- All documentation used by project participants as the basis for assumptions and source of data for establishing the baseline scenario is correctly quoted and interpreted in the Joint PDMR
- The assessment team also concluded that the identified baseline scenario reasonably represents what would occur in the absence of the project activity.

3.4.5 Additionality

The additionality of the Project is demonstrated by applying the following approach, consisting of two components:

i. A Legal Requirement Test; and

ii. An Additionality Test either based on a Positive List test or a projects-specific additionality test.

The project is not mandated/enforced by law and is entirely a voluntary activity as confirmed by the VVB since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement.

To demonstrate additionality test, the applied methodology requires the project developer to determine the additionality based on “Methodological Tool- Tool for the demonstration and assessment of additionality”, Version 7.0.0. The tool provides a step-wise approach to demonstrate and assess the additionality of a project which are:

(a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;

(b) Step 1 Identification of alternatives to the project activity;

(c) Step 2 Investment analysis;

(d) Step 3 Barriers analysis; and

(e) Step 4 Common practice analysis

PD has demonstrated the step-wise approach to establish additionality of the project activity and same is discussed below;

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The project activity is a solar power project undertaken in Thoothukudi district, Tamil Nadu, India. This is not the first such project to be installed in the country or in the state and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity: Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity.

The project activity is aimed to generate power from the solar power plant using renewable energy (solar) resources and wheeling it to identified user where as in the baseline those were using grid. VVB has confirmed the same based on submitted PPA copy and energy wheeling agreement copy. Hence, the following alternatives are considered by PP which are appropriate:

Alternative 1: The proposed project activity undertaken without being registered as a Proposed project activity.

The option of project activity being undertaken without being registered as a project activity is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as the project activity faces significant financial barriers which is established below. In absence of the project activity, the end users would have used electricity from the grid through India's National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of Plant and the country.

Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid).

In such case the project proponent would have continued without investment in project activity and continued with its usual business activities. In such case, grid would continue with the fossil fuel-based power projects[27] and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: Identified realistic and credible alternative scenario (s) to the project activity.

[27] https://powermin.gov.in/en/content/power-sector-glance-all-india

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is "the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system Version 7.0"

Sub-step 1b: Consistency with mandatory laws and regulations: The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.

VVB confirmed that installation of a solar power project including the one as project activity is consistent with mandatory laws and regulations at national (India) and sub-national level (state of Tamil Nadu). The environmental regulations, legislations and policy guidelines in respect to the project activity is governed by Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi, supported by Central Pollution Control Board (CPCB). In accordance to Annexure-II MOEF&CC, OM on J-11013/41/2006-IA. II (I) dated on 07-July-2017, Solar Photovoltaic Power Projects are not covered under the ambit of EIA Notification, 2006 and its subsequent amendments and hence, it does not require preparation of EIA and pursuing Environmental Clearance from MoEF&CC.

Moreover, Solar Power Project is considered under "white category" for Consent to Establish/Operate as per regulation, and thus there is no necessity of obtaining the Consent to Operate''. Since project activity falls under white category and the non-polluting in nature, the project fulfils the compliance to the local environmental laws and regulations.

Overall, the project activity is consistent with the following mandatory laws and regulations as confirmed by the assessment team.

- Electricity Act, 2003 and it's subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003.

- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021.
- E-waste (Management and Handling) Rules, 2020.
- Plastics Waste Management Rules, 2016 amended in 2021.
- Batteries (management and handling) rules 2019

Outcome of Sub-step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations.

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 "Continuation of the current situation" has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

PD has conducted Investment analysis to determine whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in line with the Tools for sections as per "Investment Analysis" (Ver 12.0)

Sub-step 2a: Determine appropriate analysis method

There are three options for the determination of analysis method which are:

I. Simple Cost Analysis

II. Investment Comparison Analysis and

III. Benchmark Analysis

The Project activity envisages to export the power to Indian grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA) signed between PD and the end users i. e. Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity.

In the absence of the project activity, grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the project proponent was carried out before the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the "Tool for demonstration and assessment of Additionality", Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 12.0[28] specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = 9.77%

The Required return on equity (benchmark) was computed in the following manner by the PD which is acceptable:

Nominal Benchmark = {(1+Real Benchmark) *(1+Inflation rate)}-1

Where:

Default value for Real Benchmark = 9.77% (as per Appendix of CDM TOOL 27: Investment Analysis)

Inflation Rate forecast for by Reserve Bank of India (RBI) (i.e. Central Bank of India) for India.

Benchmark estimation:

[28] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf

Project	Date of Investment Decision	Commissioning date
Floating Solar project of Greenam Energy	23-January-2018[29/12/]	12-January-2022

Inflation Rate (at the time of investment decision making)

Inflation forecast (WPI Mean of annual forecasts of Non-food Manufactured Products for 10 years)	Document Date	Sources Link
4.00%	06-April-2017	(Table. A.9 – Annual average percentage change) https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=17458

Thus, PD has calculated benchmark as below which is appropriate.

Description	Value	Sources Link
Default Value for India as per UNFCCC guidelines	9.77%	EB 112, Annex 2, TOOL27 "Investment Analysis", Version 11.0 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf
Inflation forecast (WPI Mean of annual forecasts of Non-food Manufactured Products)	4.00%	(Table. A.9 – Annual average percentage change) https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=17458
Benchmark	14.16%	Calculated

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e. 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity. substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

VVB further checked the input parameters for the project as presented below:

[29] Confirmed based on minutes of board meeting confirming to go ahead with the project considering the carbon revenue. Sicne this is the earliest date on which PP has decided to go ahead with the project, same is found acceptable.

Technical Parameters				
Particulars	**Value**	**Unit**	**Source**	**VVB assessment**
Capacity of the plant	22	MW	DPR/13/	The plant capacity is considered from Detailed Project Report (DPR) prepared dated 05-January-2018. Since this the actual installed capacity and hence acceptable.
Plant Load factor	21.85%	%	DPR/13/	The plant load factor is considered from Detailed Project Report prepared dated 05-January-2018. The value is established based on 3rd party report and hence acceptable. The actual capacity utilization factor (CUF) or PLF for the plant form the date of commissioning is 18.3%. Thus, the value considered is acceptable.
Grid Availability	100	%	As a conservative approach	PD has considered the most conservative value and hence acceptable.
Auxiliary Consumption	0.4%	%	DPR/13/	The value is considered from Detailed Project Report prepared dated 05-January-2018. This is based on internal consumption records for the site purpose and found acceptable.
Transmission Loss & Charges (Wheeling charges)	3.91%	%	as per TNERC Order TP No:2 of 2017 dt 11.08.2017/14/	PD has used local state electricity board tariff order for wheeling changes estimation which was available at the time of project decision and hence accepted.
Annual Module degradation	0.43%	%	DPR/13/	The value is considered from Detailed Project Report prepared dated 05-January-2018. Since the value is substantiated by solar panel manufacturer also. Same is accepted.
Working days in a year	365	Days	DPR/13/	The project is grid connected solar plant and hence this most conservative value is accepted.
Working hours in a day	24	Hrs	DPR/13/	The project is grid connected solar plant and hence this

				most conservative value is accepted.
Technical Lifetime of the project	25	Years	DPR/13/	The value is considered from Detailed Project Report prepared dated 05-January-2018. Since the value is substantiated by solar panel manufacturer also. Same is accepted.

Financial Parameters				
Particulars	**Value**	**Unit**	**Reference**	**VVB assessment**
Annual O & M cost	7.4	Million INR/annum	DPR/13/, page 84	The value is considered from Detailed Project Report prepared dated 05-January-2018. The value is found acceptable as the actual value is 0.42 million INR/annum. Though it is drastically low from what was considered at the time of investment decision.
Escalation in O&M	5.00%	%	DPR/13/	The value is considered from Detailed Project Report prepared dated 05-January-2018. The value is found acceptable as the actual value is 3%.
Insurance	0.09%	%	DPR/13/	The value is considered from Detailed Project Report prepared dated 05-January-2018. Since it is in line with industry practice, same is acceptable.
Admin Expenses	0.14	Million INR/annum	Opinion of sector expert (Salary etc)	The value is considered from Detailed Project Report prepared dated 05-January-2018. Since it is in line with industry practice i. .e expert's opinion, same is acceptable.

Escalation in Admin Expenses from 2nd year onwards	5.00%	%	Opinion of sector expert (Salary etc)	The value is considered from Detailed Project Report prepared dated 05-January-2018. Since it is in line with industry practice i. .e expert's opinion, same is acceptable.
Inverter warranty cost from 5th year	0.78	Million	Opinion of sector expert	The value is considered from Detailed Project Report prepared dated 05-January-2018. Since it is in line with industry practice i. .e expert's opinion, same is acceptable.
Project cost	1,274	Million INR	DPR/13/, page 87	The value is considered from Detailed Project Report prepared dated 05-January-2018 and is based on quotations received. Further the actual cost is 1,528 million INR and hence acceptable.
Equity @ 25% of total cost	318.50	Million INR	DPR/13/, page 87	The value is considered from Detailed Project Report prepared dated 05-January-2018 is based on industry practice and in line with tool 27 and hence acceptable.
Debt value	955.50	Million INR	DPR/13/, page 87	The value is considered from Detailed Project Report prepared dated 05-January-2018 is based on industry practice and in line with tool 27 and hence acceptable.
Residual/scrap/salvage Value	10%	%		The value is based on industry practice as confirmed based on https://cercind.gov.in/2017/regulation/Noti131.pdf and hence acceptable.

Tariff rate	4.12	INR/ kWh	DPR/13/	The value is considered from Detailed Project Report prepared dated 05-January-2018 and the actual tariff rate is also the same and hence accepted.
Depreciation (Book depreciation) on civil works	10%	%	https://taxadda.com/depreciation-rates-as-per-companies-act-2013/	
Depreciation (Book depreciation) on plant & machineries	8%	%	https://taxadda.com/depreciation-rates-as-per-companies-act-2013/	
IT depreciation on building & civil works	7.69%	%	IT depreciation rate[30]	These values are based on 3rd party government documents which were checked and found correct. Hence VVB accepted the same.
IT depreciation on Plant & Machineries	40%	%	IT depreciation rate[31]	
Corporate Tax Rate	25.17%	%	https://www.pwc.com/mu/en/services/india-desk/corporate-tax.html	
Interest rate on loan	10.66%	%	https://cercind.gov.in/2017/orders/05.pdf (page 5)	
Commercial operation date	12-January-2022		Commissioning certificate	

[30] https://search.incometaxindia.gov.in/pages/results.aspx?k=7%2E69

[31] https://search.incometaxindia.gov.in/pages/results.aspx?k=7%2E69

Outcome of Step 2:

PD has submitted excel worksheet which has calculated equity IRR based on above assumptions and same is found appropriate.

Equity IRR	Benchmark (Equity IRR)
5.17%	14.16%

VVB confirms that, this substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

Sub-step 2d: Sensitivity Analysis

Addressing Guidance under tool 27 - Investment Analysis, version-12.0, PD has subjected following factors to sensitivity analysis:

1. PLF value
2. O&M Cost
3. Project Cost
4. Tariff rate

Sensitivity Analysis

The results of sensitivity analysis show that the even with a variation of +10% & -10% in the project cost, O&M cost, PLF and Tariff Rate; Equity IRR is significantly lower than the benchmark. Thus, VVB confirms that the project remains additional even under the most favourable conditions.

Sensitivity Parameter 1 : PLF value
PLF considered in financials for is as per Third Party report in line with “Guidelines for the reporting and validation of Plant load factors” stated in EB 48 Annex 11 option 3(b) with a value of 21%. VVB has checked the actual PLF data also from date of commissioning till date and the vale is 18.3%. Thus, variation in PLF of more than 10% is unlikely to happen as the PLF has been reported as per the Third Party Report based on long term data. As confirmed earlier, PLF value is not increasing than the value considered at the time of decision
Sensitivity Parameter 2: O&M cost
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely as VVB has confirmed that PD has signed and O & M agreement with 3% escalation in each year.
Sensitivity Parameter 3 : Project Cost
Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached.

Based on the actual project cost, VVB noted that actual project is higher than the estimated one. The assessment is presented in the above table.
Sensitivity Parameter 4 : Tariff Rate
The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, PD has conducted sensitivity for +/-10% even then the benchmark is not breached.

Result of Sensitivity Analysis Table:

Parameters	-10%	IRR (Base Case)	10%
PLF value	1.33%	5.17%	9.01%
O&M cost	5.46%	5.17%	4.87%
Project Cost	8.98%	5.17%	2.12%
Tariff	1.33%	5.17%	9.01%

Demonstration of Additionality with Actual IRR

Since the project activity is already commissioned and operating for more than one year. PD has calculated Project activity IRR based on actual PLF, project cost, O&M and Tariff.

Actual IRR after considering the actual values for main four input parameters is coming 1.95% which less than the benchmark, hence project activity is deemed additional in actual scenario.

Step 3: Barrier Analysis

As per Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 or Step 3 or both can be used to demonstrate additionality of the project activity. In this case, PD has not used step 3.

Step 4: Common Practice Analysis

As per para 57 of Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 analysis shall be complemented with an analysis of extent to which the proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region. This test is a credibility check to complement the investment analysis.

Common Practice Analysis – 22 MW Floating PV Solar Power Project at Tamil Nadu by Greenam Energy pvt Ltd

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Since, the project capacity is 22 MW(AC), PD has selected range of solar plants between 11 to 33 MW (AC).

Range	Capacity	Unit
+50%	33	MW
Capacity of the project activity	22	MW
-50%	11	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;
- The projects apply the same measure as the proposed project activity;
- The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;
- The capacity or output of the projects is within the applicable capacity range for the chosen projects.
- The projects started commercial operation before the PD is published for global stakeholder consultation or before the start date of project activity, whichever is earlier for the project activity.

PD has carried out identification of the similar projects (CDM and non-CDM) as per sub-steps of Step (2) as follows:

- As the project is located in Tamil Nadu state of India, therefore, projects in the geographical area of Tamil Nadu have been chosen for analysis. This is found acceptable as the generated electricity is being sold to the end users are who are located within the same geographical region i. e. Tamil Nadu.

  The project activity involves generation of electricity from solar energy. The project activity is located in Tamil Nadu and the policy applicable for the solar projects is regulated by respective state policy. The policies/tariff for each state is regulated by State Electricity Regulatory Commissions of respective states and they differ for respective states. The project is Floating PV solar and a 30% equity-based project and generated power will be used by identified user through grid and paying wheeling charges.

- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are selected by PD.
- The energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.
- The project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 11 MW to 33 MW.
- Purchase Order date of the project activity is 28-August-2018, therefore, projects, which have started commercial operation before 28-August-2018, have been considered for analysis.

VVB noted that all the above information is correct and is sourced from government of India databased and hence acceptable.

Numbers of similar projects identified, which fulfil above-mentioned conditioned are N_{solar} = 0

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, Nall.

Project activities, which have got registered or are under validation with CDM/VCS/GS/GCC have been excluded in this step. Since N_{solar} = 0

N_{all} = 0[32]

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}.

As per the tool on Common Practice, the project activities have been separated from the different technologies on the basis two criteria:

Different technologies - are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed project activity and applicable geographical area):

(a) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);

(b) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);

(c) Size of installation (power capacity)/energy savings:

(i) Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6); (ii) Small (as defined in paragraph 28 of decision 1/CMP.2);

(iii) Large.

(d) Investment climate on the date of the investment decision, inter alia:

(i) Access to technology;

(ii) Subsidies or other financial flows;

(iii) Promotional policies;

(iv) Legal regulations.

(e) Other features, inter alia:

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[32] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

(i) Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

The Project is 30% equity-based and Floating PV solar grid connected project and deployed for export of power after paying wheeling charges. Hence, the investment climate of the project is different from any other project of different tariff or revenue structure. So solar power project that are not covered under reverse bid project are considered as different from the project activity.

Hence, projects where either of the conditions is satisfied those projects are counted for calculating

N_{diff} projects.

$N_{diff} = 0$[33]

Step (5): calculate factor $F = 1 - N_{diff} / N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

The factor for the project activity is calculated as follows:

In terms of the formula: $F = 1 - N_{diff} / N_{all}$, the result is mathematically meaningless.

$N_{all} - N_{diff} = 0$[34]

Thus, the project does not fulfil the requirement that F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3, and the project is not a common practice

VVB confirms that the project's additionality is well justified in the JPD&MR which was checked based on supporting evidences. The assumptions and values considered to demonstrate additionality were found reasonable and correct. Hence, VVB confirms that the project is additional and additionality is well demonstrated in line with applicable methodology and tools.

3.4.6 Quantification of GHG Emission Reductions and Removals

According to paragraph 39 of applied Methodology, *Baseline emissions include only CO2 emissions from electricity generation either in grid connected and or captive fossil fuel fired power plants that are displaced due to the project activity*. The baseline emissions are calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{CO2,y} + EG_{PJsurplus,y} \times EF_{grid,y} \quad \text{Equation (6)}$$

Where,

[33] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

[34] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

BE_y = Baseline emissions in year y (t CO_2/yr)
$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and supplied to a captive consumer as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{PJsurplus,y}$ = Quantity of the surplus electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{CO2,y}$ = The electricity emission factor (t CO_2/MWh) of the baseline electricity source
$EF_{grid,y}$ = The emission factor of the grid (t CO_2/MWh) following the procedures described in TOOL07

For the current project activity, VVB has confirmed that the 100% of generated electricity will be consumed by captive users and hence parameter – $EG_{PJsurplus,y}$ is not applicable. Moreover, electricity from grid is the baseline for this project and hence $EF_{CO2,y}$ = $EF_{grid,y}$ which is acceptable.

Thus, baseline emission would be

$$BE_y = EG_{PJ,y} \times EF_{CO2,y}$$

As per paragraph 41 of the applied methodology,

$$EG_{PJ,y} = EG_{facility,y} \qquad \text{Equation (7)}$$

Where,

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and supplied to a captive consumer as a result of the implementation of the project activity in year y (MWh/yr)
$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the captive consumer in year y (MWh/yr)

Determination of CO_2 emission factor of grid ($EF_{CO2,y}$ = $EF_{grid,CM,y}$):

Option (a) under paragraph 40 of the methodology document is chosen to calculate the CO_2 emission factor of grid. The option (a) under paragraph 40 of the methodology document states as follows:

In cases where the baseline is use of electricity from the grid (baseline scenario S1), follow the procedures described in TOOL07 to calculate the electricity emission factor.

Applicability of the tool 07: The tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity, i.e., where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).

The project activity supplies electricity to Indian Electricity grid for further captive usage. Hence this tool is applicable in case of project activity.

operating margin (OM) method

The weighted average CM method (option A) should be used as the preferred option. The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EF_{grid,OM,y}$	=	Operating margin CO_2 emission factor in year y (tCO_2/MWh)
W_{OM}	=	Weighting of operating margin emissions factor (%)
w_{BM}	=	Weighting of build margin emissions factor (%)

The following default values should be used for w_{OM} and w_{BM}:

Wind and solar power generation project activities: w_{OM} = 0.75 and w_{BM} = 0.25 (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods.

All other projects: w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period, and w_{OM} = 0.25 and w_{BM} = 0.75 for the second and third crediting period, unless otherwise specified in the approved methodology which refers to this tool.

As mentioned before, the baseline emission factors have been calculated as per CEA sourced data for Indian grid according to the formulas specified above. As this is the most authentic information available in the public domain, the baseline emission factor used in the calculation of baseline emissions for the proposed project activity is being referred from the same for transparency and conservativeness[35]

$EF_{grid,CM,y}$ = (0.9518 *0.75) +(0.8687 *0.25)

$EF_{grid,CM,y}$ = 0.9310 tCO_2/MWh

Project emissions:

As per para 25 of the applied methodology, For most renewable energy power generation project activities, PEy = 0

Therefore, the project emissions, PE_y = 0.

[35] http://www.cea.nic.in/reports/planning/cdm_co2/cdm_co2.htm

Leakage emissions:

As per para 42 of the applied methodology, No other leakage emissions are considered. Therefore, $LE_y = 0$.

Emission reductions are calculated (as per para 62 and Equation 43 of AM0123, version 1.0:

$ER_Y = BE_Y - PE_Y$

Where:

ER_y = Emission reductions in year y (tCO_2/y)

BE_y = Baseline emissions in year y (tCO_2/y)

PE_y = Project emissions in year y (tCO_2/y)

The vintage wise annual emission reductions from the project are as below:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
Year 1 (12-January-2022 to 31-December-2022)	37,614	0	0	37,614
Year 2 (01-January-2023 to 31-December-2023)	38,726	0	0	38,726
Year 3 (01-January-2024 to 31-December-2024)	38,559	0	0	38,559
Year 4 (01-January-2025 to 31-December-2025)	38,394	0	0	38,394
Year 5 (01-January-2026 to 31-December-2026)	38,229	0	0	38,229
Year 6 (01-January-2027 to 31-December-2027)	38,064	0	0	38,064
Year 7.a from (01-January-2028 to 31-December-2028)	37,900	0	0	37,900
Year 7.b from (01-January-2029 to 11-January-2029)	1,142	0	0	1,142
Total	**268,629**	**0**	**0**	**268,629**

The emission reductions for 07 years of crediting period are presented below from the project start date:

Year	Estimated baseline emissions or removals (tCO2e)	Estimated project emissions or removals (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated net GHG emission reductions or removals (tCO2e)
Year 1 (12-January-2022 to 11-January-2023)	38,893	0	0	38,893
Year 2 (12-January-2023 to 11-January-2024)	38,726	0	0	38,726
Year 3 (12-January-2024 to 11-January-2025)	38,559	0	0	38,559
Year 4 (12-January-2025 to 11-January-2026)	38,394	0	0	38,394
Year 5 (12-January-2026 to 11-January-2027)	38,229	0	0	38,229
Year 6 (12-January-2027 to 11-January-2028)	38,064	0	0	38,064
Year 7 (12-January-2028 to 11-January-2029)	37,900	0	0	37,900
Total	**268,765**	**0**	**0**	**268,765**

VVB confirms that

- all relevant assumptions and data are listed in the final joint PDMR, including their references and sources which is checked and confirmed. All data and parameter values used in the project description are considered reasonable in the context of the project.
- All estimates of the baseline emissions can be replicated using the data and parameter values provided in the joint PDMR and are calculations provided in the supporting emission reduction workbook are found correct.

3.4.7

VVB confirms that the methodology and referenced tools have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals during the project crediting period and for the current monitoring period.

Methodology Deviations

Assessment team confirms that no methodology deviation is applicable for the present project activity.

Monitoring Plan

3.4.8

The only relevant data that has to be monitored is the net electricity generation ($EG_{PJ,y}$) per year. The monitoring plan for the project activity has been developed in accordance with applicable methodology AM0123, version 1.0. The methodology requires monitoring of the quantity of net electricity generation from the proposed project activity supplied to the grid.

The assessment team has checked during the on-site visit that, the main and check meters are installed at the metering point at the project site. These meters readings are automatically stored in meter (Billed) on every month 1st 00:00 hrs. This reading (4 digit accuracy) will be automatically uploaded to TNEB main server at their HQ. Based on these readings the Monthly Generation statement will be developed in 4 digit accuracy automatically and the same will be displayed in consumer's portal (Greenam OA portal) at TNEB website.

The calibration of all the meters is planned to be is carried out in-line with the National standard which recommends at least once in 5-year calibration. Faulty meters will be duly replaced. The meters are of accuracy class 0.2s

The project has one monitoring parameter i. e. $EG_{PJ,y}$ - Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y as a part of monitoring plan.

The Net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity is estimated as a difference of electricity exported to the grid and electricity imported from the grid. The net electricity supplied/exported to the grid can be obtained from the record of Monthly Joint Meter Reading by the state utility. Thus, Net electricity supplied to the grid by the project plant in a given month = Export, kWh – Import, kWh. This value can be cross checked against the invoices raised by the end users to the PD.

PD has assured that copies of monitored data sheet and monthly energy meter reading reports will be maintained for the entire crediting period plus two years or of the last issuance of credits for this project activity, whichever occurs later.

The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during the site visit.

During on-site visit, energy meter details were checked and confirmed against calibration certificates. The details are provided in annexure 5.

3.5 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

During the onsite assessment, it was concluded that the project is implemented as per the requirement of the VCS Joint PDMR and mentioned monitoring plan, during the current monitoring period.

The project undergone continuous operation with scheduled maintenance as per the manufactures specification and few breakdowns were observed as mention in appendix 6 below which is acceptable to the assessment team. The continual energy generation unless its shutdown/breakdown is evident from the data from the monthly meter readings and meter reading protocols for the whole of the monitoring period provided as cross-check values apart from the invoices.

The project activity is an implementation of 22 MW (AC) Solar Power Plant. The sole purpose of project activity is to produce electricity for transmission into the Indian electricity grid, same is confirmed by the verification team during the onsite assessment.

VVB has checked the data and parameters used to calculate the GHG emission reductions and removals for this verification period, and has assessed the following for each of them:

- *The accuracy of GHG emission reductions and removals, including accuracy of spreadsheet formulae, conversions and aggregations, and consistent use of the data and parameters.*
- *The appropriateness of any default values used in the monitoring report.*

VVB noted that calculated GHG emission reductions and removals for this verification period are correct and are without any manual transposition errors between data sets as confirmed based on the data from the monthly meter readings and meter reading protocols for the whole of the monitoring period provided as cross-check values apart from the invoices.

Verification team confirmed that the location of the project activity including the coordinates is same as mentioned in the VCS Joint PDMR.

Assessment team checked the commissioning certificate and confirmed that the date of Commission for the project activity is correct. Assessment team also conform during onsite visit with the PP\ representatives that there is no change in project design and the project is implemented as per the description provided in the VCS PDMR.

The project boundary includes the electricity generation equipment at the project site. Assessment team also checked the technical details of the project activity implemented from documents submitted by PD. The assessment team confirmed that there is no proposed or actual change to the project design during this monitoring period. The project design as mentioned in the VCS Joint PDMR is implemented and thus the same is acceptable to the assessment team.

All required monitoring equipment's and procedures as mentioned in the latest version of VCS Joint PDMR and are available and implemented in an appropriate manner.

The project is not participating in any other environmental credits or other GHG programs and has not been rejected by any GHG programs. The project activity is not participating in any other Emission trading programme and other binding limits. The declaration mentioning the same has been submitted by the PP to the VVB. The project has implemented activities that contribute to sustainable development as described in the Joint PDMR and the same has been verified during the onsite visit.

VVB concludes that GHG emission reductions and removals provided in the project's GHG statement have been quantified correctly in accordance with the monitoring plan and applied methodology for this verification period.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

The verification team assessed whether the data and calculations of GHG emission reductions achieved resulting from the Joint PDMR. The verification team has checked whether calculations of baseline GHG emissions, project GHG emissions and leakage GHG emissions have been carried out in accordance with the formulae and methods described in the monitoring plan of the Joint PDMR.

The baseline emissions are the product of electrical energy baseline $EG_{PJ,y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by the grid emission factor.

$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$

Where:

BE_y = Baseline Emissions in year y (t CO_2)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y

$EF_{grid,CM,y}$ = CO_2 emission factor of the grid in year y (t CO_2/MWh)

Emission reductions due to project activity is calculated according to "Tool to calculate the emission factor for an electricity system" version 07.0.

Project Emissions:

The project emissions, PE_y = 0.

Leakage Emissions:

The leakage emissions, LE_y = 0

Hence,

$BE_{y} = EG_{PJ,y} * EF_{grid,CM,y}$

Where:

BE_y: Baseline Emissions in year y (tCO_{2e})

$EG_{PJ,y}$: Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y

$EF_{grid,CM,y}$: Combined Margin Emission factor for the project activity = 0.9310 tCO_2e/MWh

Baseline emissions calculations are provided below Thus,

$BE_{y} = EG_{PJ,y} * EF_{grid,y}$

Baseline Emissions for the current monitoring period are,

= 38,855.72 MWh/year * 0.9310 tCO_2e/MWh

= 36,174 tCO_2e/year (rounded down)

Project Emissions

PE_y = 0 tCO_2e

Leakage Emissions:

LE_y =0 tCO_2e

Hence, emission reductions are,

$ER_Y = BE_Y - PE_Y$

Where:

ER_y	=	Emission reductions in year y (t CO_2/y)
BE_y	=	Baseline emissions in year y (t CO_2/y)
PE_y	=	Project emissions in year y (t CO_2/y)

$ER_y = BE_y$ = 38,855.72 MWh/year * 0.9310 tCO_2e/MWh

= 36,174 tCO_2e/year (round down value)

Thus, emission reductions for the monitoring period (12-January-2022 to 28-February-2023) of 1st crediting period are,

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
Year 1 (12-January-2022- to 31-December-2022)	30,848	0	0	30,848
Year 2 (01-January-2023 to 28-February-2023)	5,326	0	0	5,326
Total	**36,174**	**0**	**0**	**36,174**

The verification team has checked the monthly generation reports and invoices for the whole of the monitoring and found all the parameters are monitored and recorded as per the monitoring plan in the Joint PDMR.

During monitoring period, the value of electricity generation is 38,855.72 MWh as confirmed based on submitted generation records which were checked. The verification team has crosschecked the emission reduction sheet and monitoring report data with the monthly meter readings and invoice bills. While verifying the Emission Reduction, the assessment team noticed a disparity between the electricity generation values reported in the JMR and the values mentioned in the invoices. This inconsistency arises from the fact that the invoices account for discounts related to transmission charges (4.11%) and transmission losses, as per the TNERC tariff order[/14/]. Consequently, for the calculation of emission reductions, PP has considered the actual electricity fed into the grid rather than the total electricity generated. This approach has been accepted by the assessment team.

Vintage wise electricity generation and subsequent emission reductions as presented in the emission reduction sheet were checked and confirmed.

Vintage period	Electricity generation (MWh)	Emission reduction (tCO_2e)
Year 1 (12-January-2022- to 31-December-2022)	33,134.74	30,848
Year 2 (01-January-2023 to 28-February-2023)	5,720.98	5,326
Total	**38,855.72**	**36,174**

In absence of the project activity net energy generated and supplied to the grid will be utilized by the existing power grid that generates electricity predominantly from fossil fuel powered thermal power plants.

The verification team checked the break down log for the monitoring period. During the verification site visit each meter location of the Solar Power Plant is also checked. The Calibration details of the monitoring meters are also checked with calibration certificates[/15/].

The verification team checked the break down log for the monitoring period. During the physical audit, meter location of project is also checked. Quantity of net electricity displaced as a result of the implementation of this project activity is the monitoring parameter as per the Joint PDMR. The energy meters installed are of accuracy class 0.2S. Net electricity supplied to grid is calculated as the difference of the measured values of “export” and “import” at the delivery point (i.e., at Grid substation). The energy import and export readings are recorded on monthly basis in the form of Joint Meter Readings (JMR). The readings are taken in presence of representative from PD and representative from state electricity board. The JMR document is signed by all these representatives. This net electricity supplied is utilized for the purpose of GHG emission reductions by the end users.

The energy meter details like make, Serial number, Calibration dates etc. are provided in appendix 05 of this report.

No delayed calibrations were observed in the project activity for this monitoring period. All the meters are of same accuracy class i.e., 0.2s. Interview during Site visit with O&M personnel also confirms the same.

The break down log is checked and there is no major breakdown during the monitoring period. No unforced error observed. No sampling procedure applied for monitoring of the data parameter and entire documents were checked by the assessment team to arrive at positive verification conclusions. The monitoring plan is followed at the project site. The monitoring meters were calibrated in line with the registered monitoring plan and there was no delay in calibration observed. Thus, assessment team concluded that the evidences are sufficient in quantity, and appropriate for the quality, to determine the GHG reductions and removals.

Assessment team checked the calculation of Estimated VCUs (38,893 tCO_2e) vs. Achieved VCUs (36,174 tCO_2e).

Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
37,219 **For Year 1 (12-January-2022- to 31-December-2022)**	30,848	-17.12%	During the 1st vintage period of present monitoring period, the project witnessed a decrease of 17.12% in emission reductions as compared to ex-ante emissions, which is due to lower PLF achieved and is nature dependent and not in control of PP
6,203	5,326	- 14.14	During the 1st vintage period of present monitoring period, the project witnessed a decrease of

For Year 2 (01-January-2023 to 28-February-2023)			14.14% in emission reductions as compared to ex-ante emissions, which is due to lower PLF achieved and is nature dependent and not in control of PP
43,422	36,174	-16.69%	During the present monitoring period, the project witnessed a decrease of 16.69% in emission reductions as compared to ex-ante emissions, which is due to lower PLF achieved and is nature dependent and not in control of PP

The actual emission reduction has been observed to be 16.69% less (Overall project) in comparison with the ex-ante estimations due to low plant load factor (PLF) and other parameters like climatic conditions which are not in the control of project proponent. The same was confirmed with PP during onsite assessment.

5 VALIDATION AND VERIFICATION OPINION

Validation Conclusion:

Applus+ Certification has been engaged by "EKI Energy Services Limited" to perform the Joint validation and verification of the "Floating Solar project of Greenam Energy".

The management of the project participant/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project's Monitoring Plan in the Joint PDMR and the approved methodology; AM0123:, version 1.0.

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

- The projects description compliance with relevant rules, including the Host Country legislation and sustainability criteria along with VCS program guide version 4.4 and standard version 4.5.
- The project's baseline and additionality are assessed against "AM0123, version 1.0
- The project's monitoring plan is assessed against "AM0123, version 1.0.
- A risk-based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project participant have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

The project is expected to generate 268,629 tCO_2e during the length of entire 1st crediting period (12-January-2022 to 11-January-2029).

Verification Conclusion: -

Our Verification approach was based on the requirements as defined under the Kyoto Protocol, and as defined by the CDM Executive Board and VERRA. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these.

The verification can confirm that;

- The project is operated as planned and described in the project document;
- The monitoring plan is as per the applied methodology;
- The monitoring process in Monitoring Report is as per the Joint PDMR.

- The development and maintenance of records and reporting procedures are in accordance with the monitoring plan;
- The installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately
- The monitoring system is in place and generates GHG emission reductions data;
- The GHG emission reductions are calculated without material misstatements.
- No limitation observed for the present verification

Verification period: 12-January-2022 to 28-February-2023 (first and last date included)
Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
Year 1 (12-January-2022-to 31-December-2022)	30,848	0	0	30,848
Year 2 (01-January-2023 to 28-February-2023)	5,326	0	0	5,326
Total	**36,174**	0	0	**36,174**

Ex-ante emissions reductions /removals	Achieved emissions reductions /removals	Percent difference	Justification for the difference
43,422	36,174	-16.69%	During the present monitoring period, the project witnessed a decrease of 16.69% in emission reductions as compared to ex-ante emissions, which is due to lower PLF achieved and is nature dependent and not in control of PP

APPENDIX 1: DOCUMENTS REVIEW UNDER VALIDATION & VERIFICATION

No.	Author	Title	References to the document	Provider
1.	Verra	VCS listed project - "FLOATING SOLAR PROJECT OF GREENAM ENERGY"	https://registry.verra.org/app/projectDetail/VCS/4187	Project participant
2.	GEPL, SPICL, GFL	PPA between GEPL & SPICL & GFL Energy wheeling agreement between GEPL & SPICL & GFL Auditor's certificate (Certificate by Chartered accountant regarding project ownership)	16-March-2018 12-January-2022	Project participant
3.	CDM	Applied methodology, AM0123 Renewable energy generation for captive use,	Version 1.0	
4.	VCS	VCS standard version 4.5 VCS program guide version 4.4 VCS validation and verification manual version 3.2	https://verra.org/programs/verified-carbon-standard/vcs-program-details/#rules-and-requirements	Project participant
5.	VCS	Joint PD&MR Emission Reduction Calculation Sheet	Version : 06 dated 17-July-2024 Version : 2 dated 17-July-2024	PP
6.	TNPGDC	Commissioning Certificate	Date : 12-January-2022	Project participant
7.	GEPL	Purchase Order	Date : 22-November-2018	PP
8.	CDM	CDM Validation and verification standard CDM TOOL 05 "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" CDM TOOL 07 "Tool to calculate the emission factor for an electricity system (Version 07.0, EB 100, Annex 4" CDM TOOL 27: Investment Analysis Version 12 CDM TOOL 02 Combined tool to	Version 3.0 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-	CDM website

No.	Author	Title	References to the document	Provider
		identify the baseline scenario and demonstrate additionality, version 07.0	tool-02-v7.0.pdf	
9.	Jinco Solar	Technical Brochure mentioning Warranty period	N/A	PP
10.	GEPL	Declaration to avoid double counting	Dated: 28-April-2023	PP
11.	GEPL	Stakeholder meeting Invitations Stake Holder Meeting feedback Forms Stake Holder Meeting Photos	Dated :02-March-2022	PP
12.	GEPL	Minutes of meeting of Board meeting confirming project consideration of Carbon revenue	Dated: 21-August-2023 Board meeting date :23-January -2018	PP
13.	Sgurr Energy	Detailed Project report	Dated: January 2018	PP
14.	TNERC	Tariff Order	Dated : 11-August-2017	PP
15.	Global Epic India Pvt Ltd	Energy Meter Specification and Calibration certificates	26-October-2021	PP
16.	PP	Grievance Register	N/A	Project participant

APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

Table 1. CLs from this Project Verification

CL ID	01	Section no.	3.1	Date : 19-July-2023
Description of CL				
PP requested to submit the following documents to the assessment team, 1. O&M Agreement. 2. Board resolution letter 3. Common Practice Analysis Sheet 4. As per section 1.1 of VCS Joint PD MR “A common power purchase agreement (PPA) had been executed between GEPL and the off-takers, namely SPIC and GFL”. PP shall submit the same. 5. Consent to establish and consent to operate. 6. PP has mentioned investment decision is 23/01/2018 however supporting document for the same is not submitted. 7. Purchase Order date of the project activity is 28-August-2018. PP to submit the copy of PO to confirm the same. PP has used nomenclature Project participant / PO in the submitted VCS Joint PD & MR. Clarification is sought.				
Project participant response				**Date :** 24-July-2023
1. O&M Agreement is now provided. 2. Board resolution letter is now provided 3. Project is first of its kind in the range 11 MW – 33 MW floating. CEA data is attached. [36] 4. PPA is submitted with this response. 5. Consent to establish and consent to operate is not applicable to this project activity as this is white category. However PP has informed Tamid Nadu PCB while implementing the project activity. 6. Board resolution letter is now provided 7. Purchase Order is now provided				
Documentation provided by project participant				
1. O&M Agreement (Asterix Energy Pvt Ltd) 2. Board resolution letter 3. Greenam PPA with SPIC & Greenstar 4. Greenam letter to MS TNPCB 5. Consent to establish and consent to operate 6. Board resolution letter *1)*				
VVB assessment				**Date:** 06-September-2023

[36] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

<table>
<tr><td colspan="2">PP has submitted the following documents:

1. PP has submitted O & M agreement between Greenam Energy Private Limited and Asterix Energy Private Limited dated 01/08/2022 for 22 MW AC (25.33 DC) floating solar plant. As per the agreement, the O & M cost is 5.1 million per year with 3% escalation per year. This value is less than that considered at the time investment decision based on submitted Detailed Project report however, has no impact on additionality of the project.
2) However, the agreement mentions plant capacity as 25.33 MWDC i.e. 25.33 MWp whereas joint PDMR mentions plant capacity as 24.7 MWp. Clarification is sought hence this part of CL is **open**.
2. The said document is not part of the submitted documents to VVB. Hence this part of CL is **open**.
3. PP has said that the project is 1st of its kind in the range for common practice analysis. The submitted weblink refers to document by Central Electricity Authority, India. PP shall be specific about the source as the said document was checked and it does not mention that the current project is 1st of its kind. Moreover, Greenam energy project with 22 MW AC was not found in the submitted list. Hence this part of CL is **open.**
4. PP has submitted copy of PPA signed between GEPL and the off-takers, namely SPIC and GFL dated 16/03/2018 . As per paragraph 10.2 the power injected will be apportioned between 3 consumers namely SPIC, GFL and TFL. However, TFL is mentioned in the VCS Joint PDMR. Hence this part of CL is **open**.
5. PP response is accepted since as per Ministry of Environment, Forests and Climate Change (MoEF&CC) such projects do no need the same. PP has submitted copy of information letter shared with local pollution control board dated 19/03/2018 for 24.7 MWp capacity which is checked and confirmed. Hence this part of CL is closed.
6. The said document is not part of the submitted documents to VVB. Hence this part of CL is **open**.
7. PP has submitted copies of purchase orders placed on various suppliers which were checked to confirm the purchase order date and same was found acceptable. Hence this part of CL is closed.
8. PP has not submitted response to the last point and a review of latest Joint PDMR has confirmed that document still mentions project participant/PO which is incorrect. Hence this part of CL is **open**.

CL 01 is open.</td></tr>
<tr><td>**Project participant response**</td><td>**Date 08/09/2023**</td></tr>
<tr><td colspan="2">1. 1. The O&M cost is for entire plant. The AC capacity is 22 MW in O&M at 1st page which is consistent in all the documents. The inconsistence in MWp is due to clerical error.
2. 2. Bord resolution is submitted now
3. 3. PP dose not claim it to be first of its kind. However in this capacity range there is no other solar project in operation before start date of project activity.
4. 4. Initially it was planned to export power to All three parties. However TANGEDCO only approved two parties and SPIC and GFL. The same can be cross checked from Wheeling agreement
5. 6. Bord resolution is submitted now
6. 8. PO is replaced with project proponent.</td></tr>
<tr><td colspan="2">**Documentation provided by project participant**</td></tr>
<tr><td colspan="2">Greenam Board Resolution dated 23rd January 2018
Greenam - Energy Wheeling Agreement signed</td></tr>
<tr><td>**VVB assessment**</td><td>**Date:** 15-September-2023</td></tr>
</table>

1. PD response is accepted. Further, during on-site visit, VVB has checked and confirmed technical specifications and energy generation records which match with the installed capacity as mentioned in the JPD-MR and hence acceptable. Hence this part of CL is closed. 2. PD has now submitted abstract of board resolution dated 23-January-2018 which confirms carbon funds were considered before going ahead for the project and the decision was based on detailed project report prepared by SgurrEnergy India dated 05-January-2018. This part of CL is closed. 3. PP has mentioned in the revised VCS PDMR that this type of project with commissioned capacity is not the first in the country and based on submitted weblink, VVB confirmed that at the time of project commissioning there were no other projects within the range of 11 MW to 33 MW which was checked and found acceptable. Hence this part of CL is closed. 4. PD has submitted wheeling agreement with TNEB dated 12-January-2022 between PD and TANGEDCO for wheeling the generated electricity for captive use of SPLC Ltd. and GFL which is valid for 25 years. Hence this part of CL is closed. 5. Closed. 6. Closed as a part of CL part 2 above. 7. Closed. 8. The revised PDMR now correctly mentioned project proponent which is checked and found okay. Hence this part of CL is closed. 9. Additional CL: Section 3.4 mentions PO date as 28-August-2018 while section 4.1 mentions Po date as 12-Match-2020. 10. Section 4.1 mentions commissioning date for energy meter. PP shall provide commissioning date for the project activity also. CL is open.	
Project participant response	**Date: 18-September-2023**
1. 9. PO date is made consistent as per PO. It is 28-Aug-18 2. 10. Project activity is also commissioned on 12-January -2022. It is corrected in revised PDMR	
Documentation provided by project participant	
Purchase order PDMR	
VVB assessment	**Date:** 16-October-2023
9. PO date is now made consistent in the revised JPDMR section 4.1 which is checked and confirmed. Hence this part of CL is closed. 10. Section 4.1 of JPDMR is revised and was checked to confirm the same. Hence this part of CL is closed. CL 01 is closed.	

CL ID	02	**Section no.**	3.1	**Date :** 19-July-2023
Description of CL				
PP requested to clarify about the project is implemented for GHG reduction purpose or not in section 1.13 of PD & MR. Section 1.14 mentions various rules and regulations. PP to provide weblinks for all of these rules and regulations.				
Project participant response				**Date :** 24-July-2023
The section 1.13 is updated as per the requirement. The weblinks for all of these rules and regulations in the Section 1.14. is now provided				
Documentation provided by project participant				
Revised Joint PD MR				
VVB assessment				**Date:** 06-September-2023

Section 1.13 of revised PDMR was checked to confirm that necessary information is now included which is appropriate and hence accepted. This part of CL is closed. Section 1.14 of revised PDMR was checked to confirm that weblinks for the rules and regulations are provided and are working. Hence this part of CL is closed. **CL 02 is closed.**

CL ID	03	**Section no.**	3.4.1	**Date :** 19-July-2023
Description of CL				
PP has applied CDM methodology ACM0002 version 21. As per EB response to AM_CLA_0304 (AM_CLA_0304), "the Board requested the MP to revise "ACM0002: Renewable energy generation for captive use", or develop a new methodology to clarify the applicability and requirements in relation to wheeling and banking of renewable electricity fed into the grid by the project activity before its use, and submit the revised methodology or new methodology, together with the revised response to the request for clarification AM_CLA_0297, for consideration by the Board at a future meeting. The Board also requested the secretariat to continue to apply the current practice to project assessment in relation to wheeling and banking. Therefore, please note that the current status of the request for clarification "AM_CLA_0297" as indicated https://cdm.unfccc.int/methodologies/PAmethodologies/clarifications/pending, is 'The EB requested the Meth Panel to review the MP87 response'.". PP to share the EB report/clarification which approved use of electricity through captive use using grid for wheeling and banking under ACM0002 methodology.				
Project participant response				**Date :** 19-July-2023
Project activity is not for captive use. All power is sold to identified user through grid. GEPL is registered company as per company act and this company is only generating power for Solar plant. This is no captive plant of this company. GEPL is selling power to SPIC and GFL , PPA is established already. Before GEPL power, SPIC and GFL were using grid as a source of power.				
Documentation provided by project participant				
PPA				
VVB assessment				**Date:** 06-September-2023
PP has now revised PDMR to mention that the project activity involves generation of solar energy and subsequent sell of the same to identified users through electricity grid. Thus, project adheres to ACM0002 applicability which is checked and confirmed based on copy of power purchase agreement. **CL 03 is closed.**				

CL ID	04	**Section no.**	3.4.2	**Date :** 19-July-2023
Description of CL				
1. PP is requested to submit an undertaking for no double accounting for current monitoring period of project activity for participation in other GHG program, 2. As per Joint PD & MR, Project activity was already commissioned. However, to verify the same PP has not provided the commissioning certificate. 3. PP request to submit Breakdown details in VCS PD MR if any, and to verify the same, PP requested to provide breakdown log sheet.				

Project participant response	**Date :** 24-July-2023
1. An undertaking for no double accounting for current monitoring period of project activity for participation in other GHG program is now provided. 2. Commissioning certificate is now provided to the verification team. 3. Breakdown log sheet is now provided to the verification team.	
Documentation provided by project participant	
1. Declaration for no double accounting 2. Commissioning certificate 3. Revised Joint PD MR	
VVB assessment	**Date:** 06-September-2023
1. PP has submitted undertaking dated 28/04/2023 which was checked and found appropriate. Hence this part of CL is closed. 2. PP has submitted copy of commissioning certificate dated 12-January-2022 which confirms the date of commissioning as mentioned in the PDMR and hence this part of CL is closed. 3. PP response is unclear. The revised PDMR was checked and it does not provide breakdown details in the same. Moreover, breakdown log sheet was not found in the submitted documents. Hence this part of CL is **open**. **CL is open.**	
Project participant response	**Date :** 08-September -2023
Breakdown detail is provided in PDMR section 4.1	
Documentation provided by project participant	
Breakdown log sheet	
VVB assessment	**Date:** 15-September-2023
PP has revised JPD-MR to include breakdown details in section 4.1 which is checked and acceptable as confirmed based on breakdown log sheet submitted. Hence this CL is closed. CL 04 is closed.	

CL ID	05	**Section no.**	3.1	**Date :** 19-July-2023
Description of CL				
In table 1 under section 1.17, PP has selected 4 SDGs namely – 7.2, 8.5 , 9.4, 13.2. PP shall mention "Current Project Contributions" for each SDG and also quantify the same. For "Contributions Over Project Lifetime", PP shall provide brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime. SDG 9.4 is "By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities". The current project is a greenfield project and is not upgrading any infrastructure or retrofitting industries. It is not clear why PP has selected this SDG as a part of monitoring.				
Project participant response				**Date :** 24-July-2023
In table 1 under section 1.17, the Current Project Contributions for each SDG and quantification of the same has been updated as per the requirement. In table 1 under section 1.17, The SDGs has been revised in the respective section.				

<table>
<tr><th colspan="2">Documentation provided by project participant</th></tr>
<tr><td colspan="2">Revised Joint PD MR</td></tr>
<tr><th>VVB assessment</th><th>Date: 06-September-2023</th></tr>
<tr><td colspan="2">Revised table in section 1.17 was checked by VVB and it is checked and confirmed that PP has now mentioned appropriate information. SDG goal 9.4 is now removed as same is not applicable which is appropriate and hence accepted.

CL 05 is closed.</td></tr>
</table>

Table 2. CARs from this Joint Validation and Verification

<table>
<tr><th>CAR ID</th><td>01</td><th>Section no.</th><td>3.1</td><td>Date : 19-July-2023</td></tr>
<tr><th colspan="5">Description of CAR</th></tr>
<tr><td colspan="5">1. PP has not provided the description of scenario existing prior to the implementation in section 1.1 as per requirement of PD & MR template version 4.2.
2. PP has not provided the required data in tabular format in section 1.1 as per requirement of PD & MR template version 4.2.
3. PP has not described and justified how the project is eligible under the scope of the VCS Program as per para 2.1.1 of VCS standard 4.2 in section 1.3 as per requirement of PD & MR template version 4.2.
4. In section 1.4 of JOINT PD & MR, “indicate whether the project has been designed to include a single location or installation only, multiple locations or project activity instances, but is not being developed as a grouped project, or as a grouped project” as per requirement of PD & MR template version 4.2.
5. PP has not provided the all required information as per PD & MR template i.e. telephone number of Project Proponent in section 1.5 of PD & MR.
6. PP has not provided the justification of start date as per PD & MR template for Project activity in section 1.8 of PD & MR.
7. PP has revise stion 1.10 as one of the options is to be chosen and not the both as per PD & MR template version 4.2.
8. In the same section i. e. 1.10, VVB observed that ER values are reducing over each year however, PP has not mentioned the reason for the same in the table or in the concerned section.
9. PP has not provided the description of energy production system, technology, equipment involved, age and average lifetime of equipment and list of facilities are missing.as per PD & MR template for Project activity in section 1.11 of PD & MR.
10. PP has mentioned Not Applicable in section 2.4 of PD & MR. As per the template requirement, “Demonstrate how due account of all and any comments received during the public comment period has been taken. Include details on any updates to the project design or demonstrate the insignificance or irrelevance of comments.”.
11. Section 4.1 shall “Describe the implementation status of the project activity(s), include information on the following: .... ” as per JOINT PD & MR template.
12. In section 5.4, as per the template year is mentioned as “Year A (DD-Month-YYYY-- DD-Month-YYYY)”. PP to follow same details and correct the column for year.
13. In section 6.3, PP shall provide the procedures used for internal auditing and QA/QC and the procedures used for handling any internal auditing performed and any non-conformances identified as per Joint PD & MR template.
14. In section 7.5, “state the estimated ex-ante GHG emission reductions and removals and the achieved emission reductions and removals for this monitoring period. Report the percentage difference and justify the difference. The quantities of GHG emission reductions and removals are the total quantities before any deductions for buffer credits.” as per the template requirement.
15. Assessment team found some inconsistencies throughout the PD & MR. Same has been commented in PD & MR.

Corrective action sought.</td></tr>
</table>

Project participant response	Date : 24-July-2023
1. The description of scenario existing prior to the implementation is now updated in section 1.1 as per requirement of PD & MR template version 4.2. 2. The required data in tabular format is now Updated in section 1.1 as per requirement of PD & MR template version 4.2. 3. The Justification of the scope of the VCS Program as per para 2.1.1 of VCS standard 4.4 is now updated in the respective section. 4. In section 1.4 of JOINT PD & MR, the information is Updated as per the requirement of template of JOINT PD & MR. 5. The telephone number of Project Proponent is now updated in section 1.5 of Joint PD & MR. 6. The justification of start date is now provided as per PD & MR template requirement for Project activity in section 1.8 of PD & MR. 7. The section 1.10 is now revised as per the requirement of joint PD & MR template version 4.2 8. ER are reducing each year due to degradation in generation 9. The description of energy production system, technology, equipment involved, age and average lifetime of equipment and list of facilities are now provided in the respective section. 10. In section 2.4 the Information is now in lined with requirement of Joint PD MR. 11. In Section 4.1 the information is now Updated as per the requirement of the Joint PD MR. 12. In section 5.4 Year A (DD-Month-YYYY-- DD-Month-YYYY) is now Updated as per the requirement of the Joint PD MR template version 4.2. 13. Statement is corrected as per practice following at the plant 14. The Information in section 7.5 in now Updated as per the requirement of the Joint PD MR template version 4.2. 15. All the sections are now consistence throughout the PD & MR as per the requirement of the Joint PD MR template version 4.2.	
Documentation provided by project participant	
Revised Joint PD MR	
VVB assessment	**Date:** 06-September-2023
1. Section 1.1 is now revised to include description of scenario prior to project implementation which is checked and found appropriate. Hence this part of CAR is closed. 2. Table in section 1.1 is now revised which is checked and confirmed. Hence this part of CAR is closed. 3. PP has not corrected PDMR in line with para 2.1.1 of VCS standard 4.2. Hence this part of CAR is **open**. 4. Section 1.4 is now revised which is checked and confirmed. Hence this part of CAR is closed. 5. Section 1.5 is now revised to include the phone number which is checked and confirmed. Hence this part of CAR is closed. 6. PP has now added justification for the project start date in section 1.8 which is checked and confirmed based on submitted copy of commissioning certificate. Hence this part of CAR is closed. 7. Section 1.10 is now revised which is checked and confirmed. Hence this part of CAR is closed. 8. PP response is accepted. However, same shall be provided in the PDMR for clarity to a 3rd party viewing the document. Hence this part of CAR is **open**. 9. PP has revised section 1.11 to provide all the required details which were checked against the submitted evidences such as technical specifications of the plant and hence acceptable. This part of CAR is closed. 10. PP has revised section 2.4 and VVB has confirmed that no public comments were received and hence accepted. This part of CAR is closed. 11. Section 4.1 is now revised to include details about implementation of the project which is checked and confirmed. Hence this part of CAR is closed. 12. Section 5.4 is now revised which is checked and confirmed. Hence this part of CAR is closed. 13. PP response is unclear. Section 6.3 shall provide the procedures used for internal auditing and QA/QC and the procedures used for handling any internal auditing performed and any non-conformances identified as per Joint PD & MR template. Hence this part of CAR is **open**. 14. Section 7.5 is now revised to include comparison and its reasoning which is checked and found appropriate. Hence this part of CAR is closed. 15. There are still few inconsistencies pending. Hence this part of CAR is **open**. **CAR 01 is open.**	
Project participant response	**Date :** 08-September - 2023

1. 3. PDMR revised and submitted as per observation, 2. 8. PDMR is updated to discuss the yearly reduction of ERs in section 5.1 3. 13. Internal audit is explained in documents. 4. 15. Inconsistency in PDMR are corrected now.	
Documentation provided by project participant	
PDMR	
VVB assessment	**Date:** 15-September-2023
1. Closed. 2. Closed. 3. PP has now corrected PDMR in line with para 2.1.1 of VCS standard 4.2 in section 1.3 which is checked and accepted. Hence this part CAR is closed. 4. Closed. 5. Closed. 6. Closed. 7. Closed. 8. PP has revised section 1.11 to include the same details which is checked and acceptable. Hence this part of CAR is closed. 9. Closed. 10. Closed. 11. Closed. 12. Closed. 13. PP has revised section 6.3 to include the same details which is checked and acceptable. Hence this part of CAR is closed 14. Closed. 15. There are still few inconsistencies pending for which comments are raised in PDMR. Hence this part of CAR is **open**.	
Project participant response	**Date :** 18-September-2023
Inconsistencies are corrected and in revised PDMR	
Documentation provided by project participant	
PDMR	
VVB assessment	**Date:** 16-October-2023
Revised JPDMR was checked and it was noted that inconsistencies are being taken care of. CAR 01 is closed .	

CAR ID	02	**Section no.**	3.1	**Date :** 19-July-2023
Description of CAR				
PP has not provided weblinks or supporting document for laws mentioned in section 1.14 of PD & MR. Corrective action sought.				
Project participant response				**Date :** 24-July-2023
The weblinks or supporting document is now provided for laws mentioned in section 1.14 of PD & MR				
Documentation provided by project participant				
Revised Joint PD MR				
VVB assessment				**Date:** 06-September-2023
Section 1.14 of revised PDMR was checked to confirm that all the necessary weblinks are provided and same are working. Hence **CAR 02 is closed**.				

CAR ID	03	Section no.	3.4.3	Date : 19-July-2023
Description of CAR				
1. In section 3.3, PP has not considered captive use of generated electricity which is wheeled through the grid. Correctio action sought. 2. Further, the current project is floating solar power plant on the existing reservoir. PP shall explain and correct why CH_4 is not included as a source of emissions as there could be anaerobic activity in these reservoirs leading to CH_4 emissions.				
Project participant response				**Date :** 24–July-2023
1. Project is not for captive use. It is exporting power to identified users. 2. Floating solar is not enhancing any activity which may lead CH4 emission. Solar panels are affixed on buoyant structures made of PVC, keeping them afloat on the water body surface.				
Documentation provided by project participant				
NA				
VVB assessment				**Date:** 06-September-2023
1. PP response is incorrect. In the project boundary diagram, PP has not shown that the generated energy is supplied to identified users. Hence this part of **CAR is open**. 2. Based on the site visit, it was confirmed that the project is a floating structure and is not enveloping the complete reservoir. Hence this part of CAR is closed. **CAR 03 is open.**				
Project participant response				**Date :**08-September -2023
1. Project boundary is corrected in revised PDMR				
Documentation provided by project participant				
PDMR				
VVB assessment				**Date:** 15-September-2023
1. PP has now corrected diagram in section 3.3 to include the end users to who electricity is being supplied which is appropriate. Hence this part of CAR is closed. CAR 03 is closed.				

CAR ID	04	Section no.	3.3.2	Date : 19-July-2023
Description of CAR				
PP has not provided the information regarding mechanism for ongoing communication with local stakeholder consultation in section 2.2 of PD & MR. Also provide the all supporting documents for local stakeholder consultation.				
Project participant response				**Date :** 24-July-2023
In section 2.2 the information regarding mechanism for ongoing communication with local stakeholder in now Updated. All the supporting documents for the same has also been submitted.				
Documentation provided by project participant				
VVB assessment				**Date:** 06-September-2023
PP has revised section 2.2 to add required details which were checked and found appropriate. Hence **CAR 04 is closed.**				

<table>
<tr><td>CAR ID</td><td>05</td><td>Section no.</td><td>3.4.5</td><td>Date : 19-July-2023</td></tr>
<tr><td colspan="5">Description of CAR</td></tr>
<tr><td colspan="5">1. Section 3.5 mentions that “The Project activity envisages to export the power to Indian grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA).”. Section 1.1 mentions that “The purpose of this Project is to utilize the area of the reservoir to generate clean energy for captive use by the AMIH Group companies - Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL). A common power purchase agreement (PPA) had been executed between GEPL and the off-takers, namely SPIC and GFL.”. Corrective action sought.
2. PP has considered tool 27 version 12 which was released on 02-November-2022. In the same section , PP has mentioned investment decision is 23/01/2018. Use of version 12 tool 27 is found incorrect.
3. PP’s investment decision date is 23/01/2018, Detailed Project Report was finalized on 16/02/2018 and project commissioning date is 12/01/2022. PP shall provide chronology of events for these 4 years till date of commissioning from decision date with supporting evidences.
4. Default value for India is mentioned 10.55% as well as 9.77%. Corrective action sought.
5. GST was implemented in India from July, 2017 however same is not considered as a part of taxation.
6. PP shall check and correct supporting documents mentioned for the values and few are inconsistent with the submitted IRR sheet.
7. Sub step 2d mentions “Addressing Guidance under Sensitivity Analysis Version-11.0”. PP shall check and correct the same.
8. Since the project is already commissioned, in the sensitivity analysis, PP shall provide assessment based on the actual values.
9. On page 34 of VCS PD & MR, it is mentioned that “The project is Floating PV solar and a 30% equity-based project and generated power will be used for captive purpose through grid and paying wheeling charges”. PP has not considered wheeling charges separately in the financial analysis.</td></tr>
<tr><td colspan="4">Project participant response</td><td>Date : 08-September -2023</td></tr>
<tr><td colspan="5">1. Project activity is exporting power to grid and selling it to Identified users. Section 3.5 is amended as per “PPA, Power Purchase Agreement (PPA) signed between Greenam Solar and Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL)”.
2. As the investment decision is 23/01/2018,Joint PDMR is revised and Version 11.0 of tool-27 (EB 112, Annex 2) has been used throughout the PDMR
3. The chronology of events from investment decision till commissioning date is now updated in the section 4.1.
4. As per PPA page no.13. all the taxes after interconnection will be borne by buyers. Tariff is exclusive of all taxes.
5. As per Tool-27 version 11.0 (EB 112, Annex 2) Default value for India is taken as 9.77%. Hence Corrected.
6. IRR sheet and supportive documents made in line
7. The statement under the section 3.5, sub step 2d. has been corrected.
8. Comparison with actuals are provided in sensitivity analysis in revised PDMR
9. Wheeling charges are taken into account in IRR calculation. As per wheeling agreement wheeling charges are given in 3.91%.</td></tr>
<tr><td colspan="5">Documentation provided by project participant</td></tr>
<tr><td colspan="5">PDMR
PPA
IRR sheet</td></tr>
<tr><td colspan="4">VVB assessment</td><td>Date: 06-September-2023</td></tr>
</table>

1. PP has revised section 3.5 which is checked and found appropriate. Hence this part of CAR is closed. 2. PP has revised version of tool 27. As per version 11 of the tool, default value is 10.55% and as per version 12, the default value is 9.77. It is not clear to VVB why a conservative value of default value as per version 12 is not used for calculating the benchmark. Hence this part of **CAR is open**. 3. PP has added chronology of events in section 4.1 of revised PDMR. The board resolution date i. e. 23-January-2018 is considered as investment decision date. The submitted Detailed Project report was prepared on 12-August-2016 and the final revision is 15-February-2018. It is evident from the submitted chronology that version B4 of DPR was available at the time of investment decision. PP shall submit copy of DPR version B$ to check and confirm the investment decision parameters considered in the financial analysis. Hence this part of **CAR is open**. 4. The assessment of this point is subject to closure of point no. 2 above hence this part of **CAR is open**. 5. PP response is unclear. 6. PP has now corrected the information in line with submitted IRR sheet which is checked and confirmed. Hence this part of CAR is closed. 7. PP has corrected the sentence under section 3.5 sub step 2d which is checked and confirmed. Hence this part of CAR is closed. 8. PP has now provided comparison of IRR value considering the actual parameters and the project is still additional. Hence this part of CAR is closed. 9. PP has not corrected the IRR sheet for wheeling charges. Moreover, wheeling agreement was signed post investment decision date. Hence this part of **CAR is open**. **CAR 05 is open.**	
Project participant response	**Date :** 08-September -2023
1. 2. Tool was applicable at the time of investment decision; decision was taken basis on the benchmark available that time. Version 12 was not published at that time. 2. 3. DPR B4 is submitted now 3. 4. Responded as above 4. Wheeling charges are applied in IRR sheet, see R.NO 13 of assumption sheet. , Wheeling charges was referred form as per TNERC order T.P.No 2 of 2017dt 11.08.2017.	
Documentation provided by project participant	
VVB assessment	**Date:** 15-September-2023
1. Closed. 2. PP has not explained why a more conservative value of benchmark as per the latest version of tool is not applied for. Hence this part of CAR is open. 3. PP has submitted DPR version B$ to check and confirm the values considered for financial analysis and were found correct. Hence this part of CAR is open. 4. Subject to closure of point 2 above. 5. GST was implemented in India from July, 2017 however same is not considered as a part of taxation. 6. Closed. 7. Closed. 8. Closed. 9. PP response is accepted and same is checked, confirmed. Hence this part of CAR is closed. CAR is open.	
Project participant response	**Date :** 18-September-2023
1. 2. Benchmark is revised as per latest available tool and which is also conservative.	
Documentation provided by project participant	
VVB assessment	**Date:** 16-October-2023

PP has now used the most conservative value of benchmark which is as per the latest version of applied tool and hence acceptable. Revised JPDMR and financial analysis sheet was checked to confirm the same and found appropriate. CAR 05 is closed.

CAR ID	06	Section no.	3.4.6	Date : 19-July-2023
Description of CAR				
As per section 5.1, $EG_{PJ,y}$ is "Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year" while in section 6.2 and 7.1 it is mentioned as "Quantity of net electricity generation supplied by the project plant/unit to the grid in year y". Corrective action sought. In section 6.2 under parameter $EG_{PJ,y}$ it is written that "*The Net electricity exported/supplied to the grid by the project activity is estimated as a difference of electricity exported to the grid and electricity imported from the grid. The net electricity supplied/exported to the grid can be obtained from the record of Monthly Joint Meter Reading by the state utility*". PP to provide cross checking mechanism for the same for better clarity.				
Project participant response				**Date :** 24-July-2023
The statement is now corrected in the respective section 6.2 and section 7.1 In section 6.2 under parameter $EG_{PJ,y}$, Calculation method, the statement is now corrected.				
Documentation provided by project participant				
VVB assessment				**Date:** 06-September-2023
PP has now corrected section 6.2 and 7.1 of revised PDMR which is checked and confirmed. Hence this part of CAR is closed. In section 6.2 of revised PDMR, PP has now added "..Thus, Net electricity supplied to the grid by the project plant in a given month = Export, kWh – Import, kWh ….". This is not a cross checking mechanism. Hence this **part of CAR is open**. **CAR 06 is open.**				
Project participant response				**Date :** 08-September -2023
Monthly generation export can be cross checked from Invoices. Cross check mechanism is updated in PDMR				
Documentation provided by project participant				
VVB assessment				**Date:** 15-September-2023
PP has now revised section 6.2 to include the cross checking mechanism which is checked and found appropriate. Hence CAR 06 is closed.				

CAR ID	07	Section no.	4.2	Date : 19-July-2023
Description of CL				
PP has not provided the information regarding actual emission reduction during current monitoring period in section 7.5 as per the PD & MR template version 4.2 requirements. Corrective action sought.				
Project participant response				**Date :** 24-July-2023
The information regarding actual emission reduction is now updated during current monitoring period in section 7.5 as per the PD & MR template version 4.2 requirements				
Documentation provided by project participant				
Revised Joint PD MR				
VVB assessment				**Date:** 15-September-2023
Section 7.5 was checked and it was confirmed that actual emission reductions are now provided. Hence CAR **07 is closed.**				

APPENDIX 3: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER

Verification Team Member:

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of DOE or outsourced entity)	Involvement in			
						Desk review	On-site inspection	Interview(s)	Verification findings
1.	Lead Auditor/ Technical Expert	OE	Pundlik	Deepak	TQC	Yes	Yes	Yes	Yes
2.	Financial Expert	OE	Singh	Jitendra Mohan	TQC	Yes	No	Yes	Yes

Technical Reviewer and Approver of the Verification and Certification Report:

No.	Role	Type of resource	Last name	First name	Affiliation (e.g., name of central or other office of DOE or outsourced entity)
1.	Technical reviewer (TR)	OE	Xue	Denny	LGAI Technological Center S.A. (Applus+ Certification)

Shorts CV of the Team:

Mr. Deepak Pundlik has experience in climate change, waste management and environmental management. After completing Masters in Environment Sciences from Pune university, He has worked in waste management field. As a GHG consultant, He handled projects under renewable energy, waste management sectors during his stint with companies - MITCON and Thermax. Post Thermax, Deepak was involved in organic farming research project with Tata Institute of Social Sciences. As a GHG auditor, He has validated/verified projects under CDM/VCS/GS and GCC mechanisms from renewable energy, energy demand, waste management sectors.

Mr. Jitendra Mohan Singh, Mr. Jitendra Mohan Singh, has done Advanced MSc in Sustainable Energy Systems and Management from International Institute of Management, University of Flensburg, Germany and B.Tech. in Agricultural Engineering from Allahabad University, India. He has more than (18) years of working experience in different organizations like IARI, IIT Delhi, ICAR, IRADe, CAPART, SMEC and Perenia Carbon and M B Power (Madhya Pradesh) Ltd. in the area of Agriculture, Energy & Environment and Climate Change. He also worked on contract basis (adhoc) as a RIT expert in UNFCCC from 2010 to 2013. Currently. e is empanelled with Applus+ Certification since 2020 and has been involved Verifications of Pas/PoAs as Lead Auditor and Technical Expert for Renewable and non-Renewable as well as Energy Demand.

Mr. Denny Xue (Master's Degree in Environmental Engineering, Bachelor's Degree in Thermal Engineering) is an Auditor appointed by Applus+ LGAI for the GHG project assessment, auditing and technical review. He has more than 6 years of work experience in CDM/GS4GG/VCS project assessment and technical review with Applus+. Before he joined Applus+ LGAI, he has been working for Shanghai Chuanji Investment and Management which is a CDM consultancy company as a project manager for CDM project development. Mr. Denny Xue is based in Shanghai, China. Mr. Denny Xue participates in the project's technical review team.

APPENDIX 4: ABBREVIATIONS

Abbreviations	Full texts
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEA	Central Electricity Authority
CER	Certified Emission Reduction(s)
CL	Clarification request
CO_2	Carbon dioxide
CO_2e	Carbon dioxide equivalent
CP	Crediting Period
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse gas(es)
GWP	Global Warming potential
SPP	Solar Power Plant
JMR	Joint Metering reading
MP	Monitoring Period
OM	Operating Margin
PD	Project Description
PP	Project Participant
TNEB	Tamil Nadu Electricity Board
VCS	Voluntary Carbon Standard
VCU	Voluntary Carbon Unit
VER	Voluntary Emissions Reductions / Verified Emissions Reductions
VVB	Validation/Verification Body

APPENDIX 5: CALIBRATION DETAILS

As per the monitoring plan, the meter calibration frequency is once in 05 years thus no delayed calibration was observed during this monitoring period.

Main meter	
Type of meter(s)	E3M024, 3Ph, 4wire
Location of meter(s)	At plant switch yard.
Accuracy of meter(s)	0.2S
Serial number of meter(s)	TNW02796 (Meters are subjected to replaced or changed as per requirement).
Calibration frequency	Once in five years
Date of Calibration/ validity	26-October-2021
Reference No. of Calibration Certificates	GEPC/2021-22/310/CC/2774
Calibration Status	Ok
Check meter	
Type of meter(s)	E3M024, 3Ph, 4wire
Location of meter(s)	At plant switch yard.
Accuracy of meter(s)	0.2S
Serial number of meter(s)	TNW02797 (Meters are subjected to replaced or changed as per requirement).
Calibration frequency	Once in five years
Date of Calibration/ validity	26-October-2021
Reference No. of Calibration Certificates	GEPC/2021-22/310/CC/2775
Calibration Status	Ok

APPENDIX 6: BREAKDOWN DETAILS

Sr. no	Date	Breakdown detail	Generation Loss (in Units)	Remarks
		FY 2021-22		
		NIL		
		FY 2022-23		
1	15-April-2022	IDT -1 Transformer got tripped in Winding temp high. Inverter 1 & 2 Generation affected for 1.15 hrs	6,000	IDT -1 taken in line after adjusting the set values in discussion with Manufacturer.
2	21-February-2023	Greenam 110 KV feeder trip due to distance protection . Plant generation affected for 10.00 hrs. From 06:30 hrs to 16:30 hrs	94,000	110 KV tower line inspected and tested by TNEB (State Power utility company) and power supply resumed.
		Total Losses	100,000	
		FY 2023 -24		
1	12-May-2023	Greenam 110 KV feeder trip due to distance protection . Plant generation affected for 7.00 hrs. From 06:15 hrs to 13:15 hrs	69,000	110 KV tower line inspected by TNEB (State Power Utility Company)and power supply resumed.